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ISO-KETOLA et al.(10) **Pub. No.: US 2025/0278159 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **A METHOD FOR OPERATING A LAYERED
DEVICE AND AN ARRANGEMENT
COMPRISING A LAYERED DEVICE**(52) **U.S. CL.**CPC **G06F 3/0447** (2019.05); **G06F 3/04166**
(2019.05); **G06F 3/0445** (2019.05); **G06F**
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ABSTRACT(73) Assignee: **FORCIOT OY**, Tampere (FI)(21) Appl. No.: **18/862,776**(22) PCT Filed: **May 2, 2023**(86) PCT No.: **PCT/FI2023/050237**

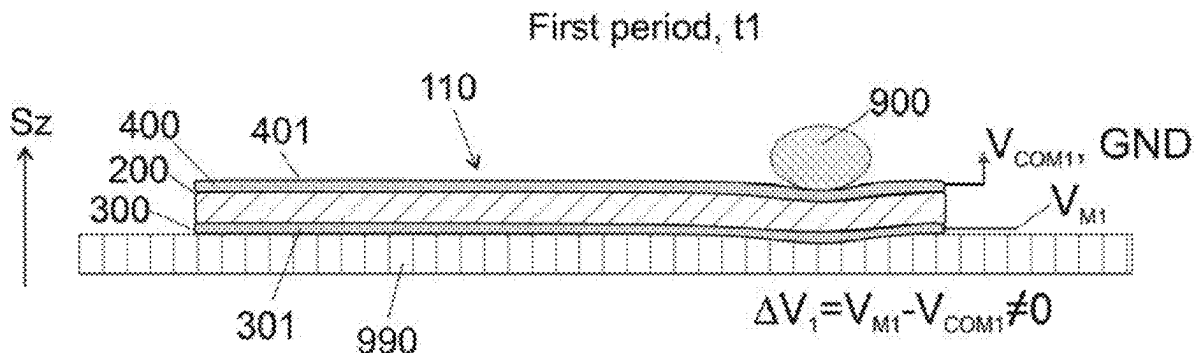
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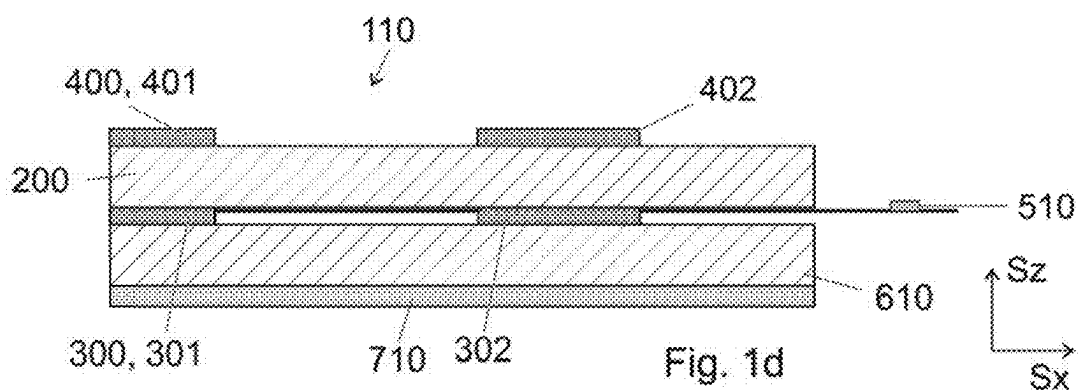
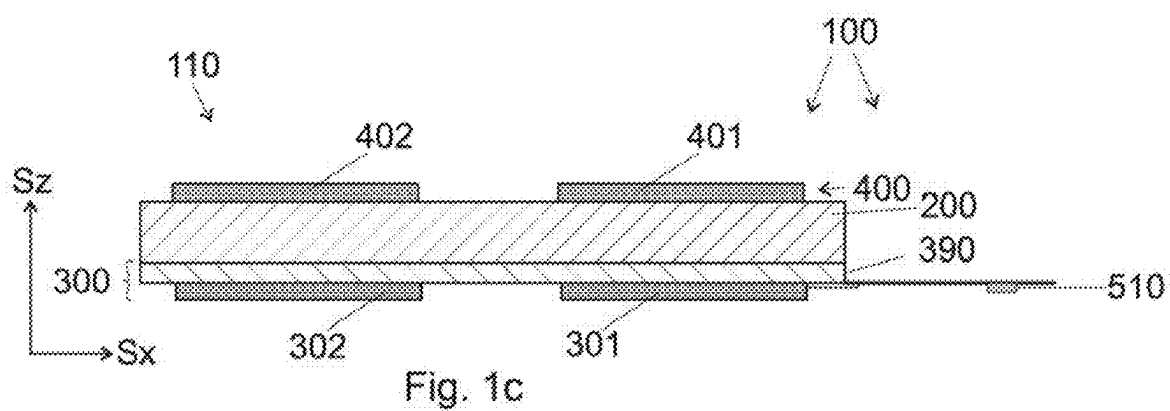
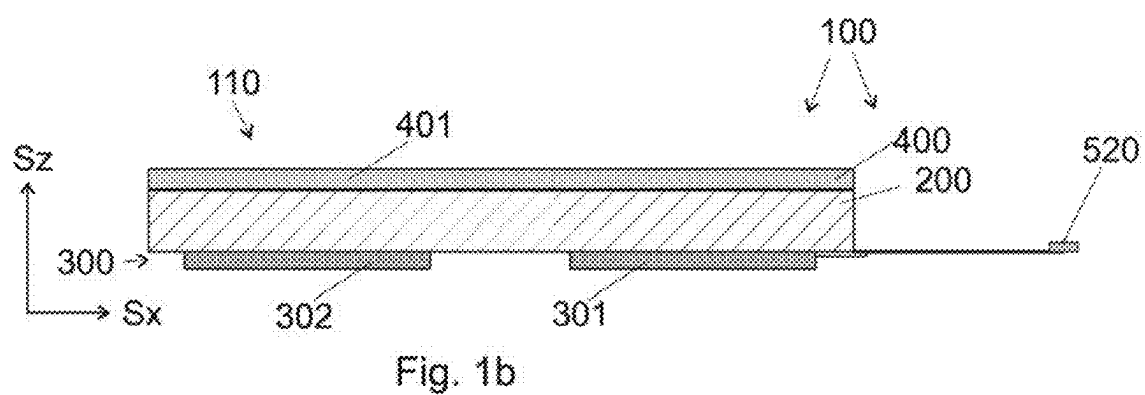
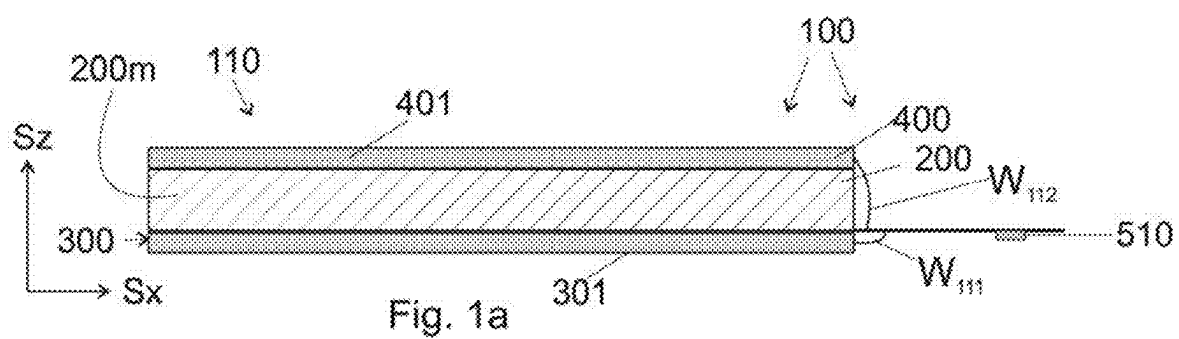
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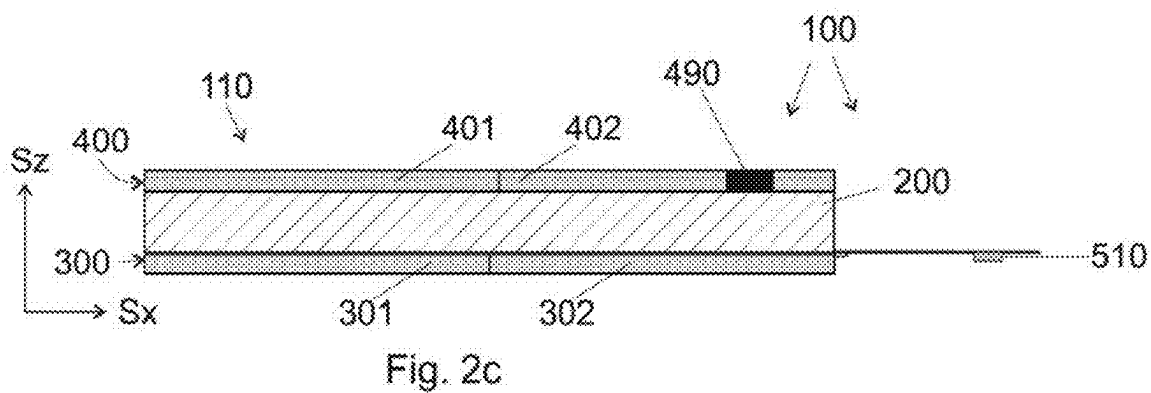
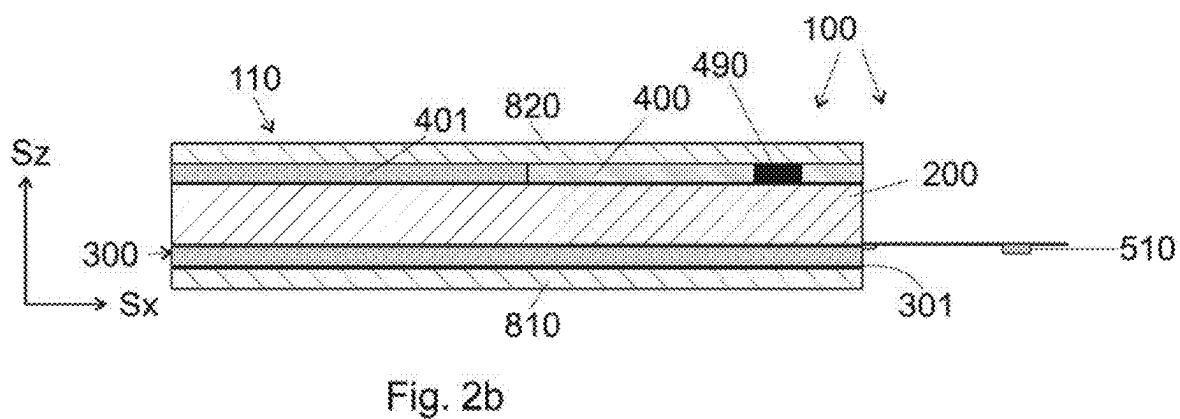
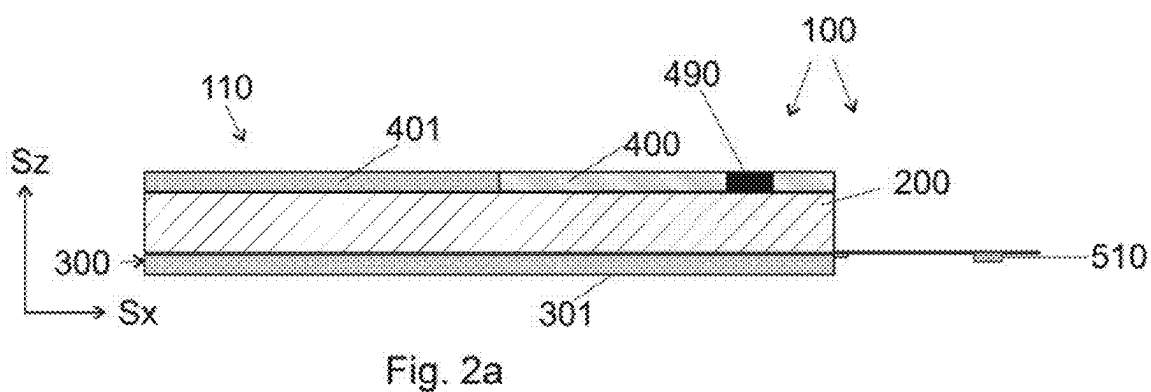
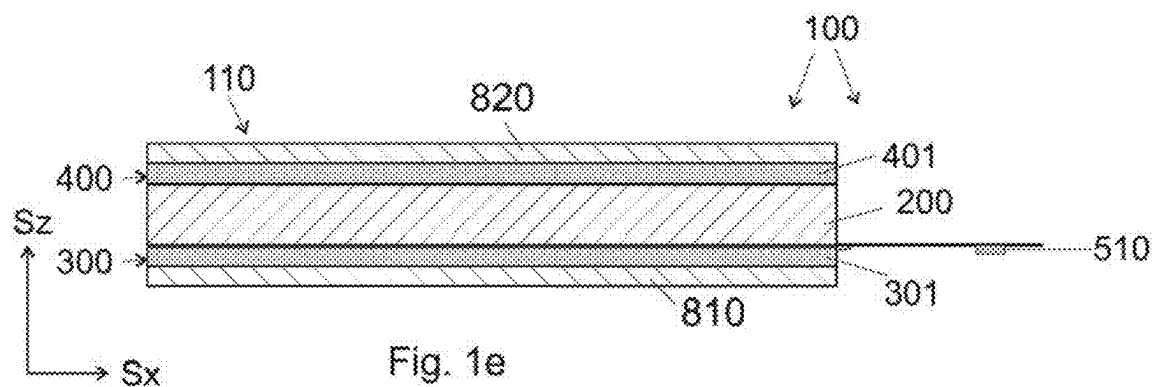
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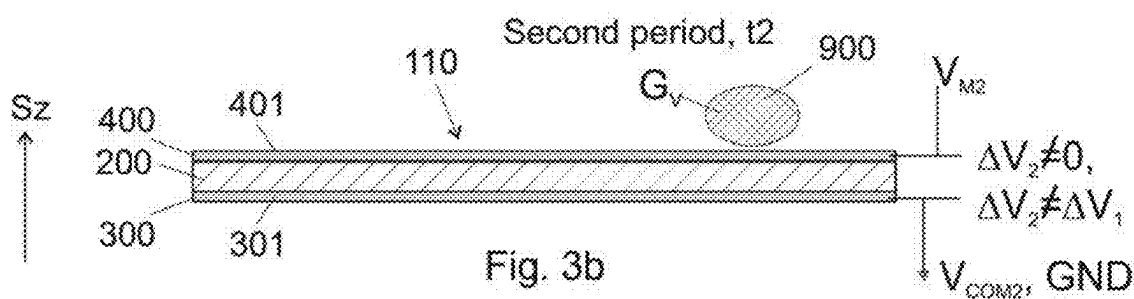
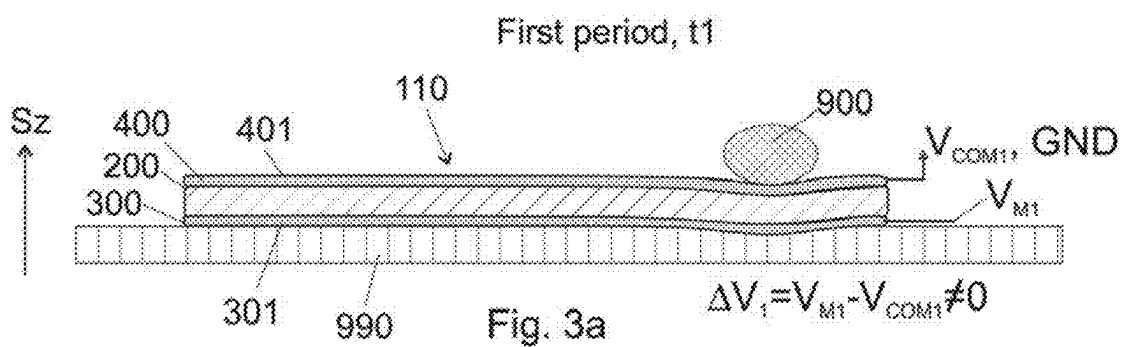
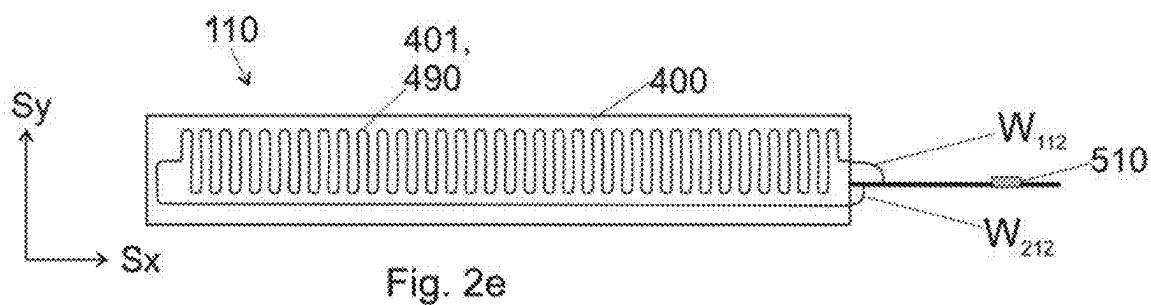
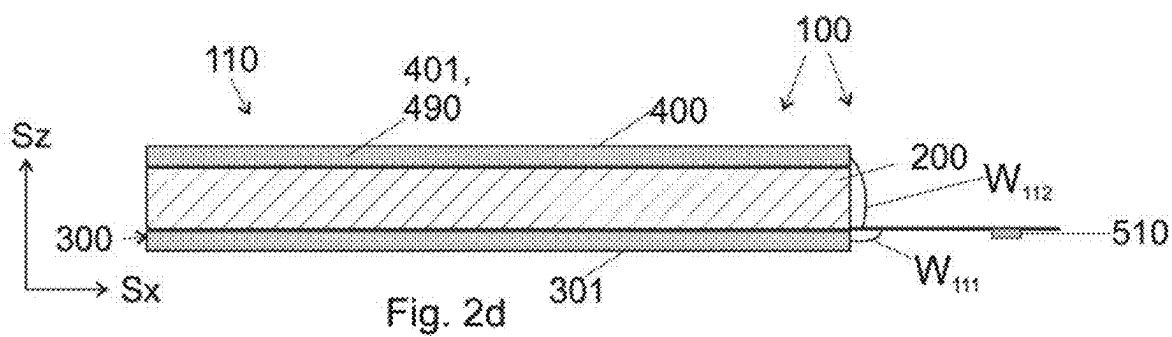
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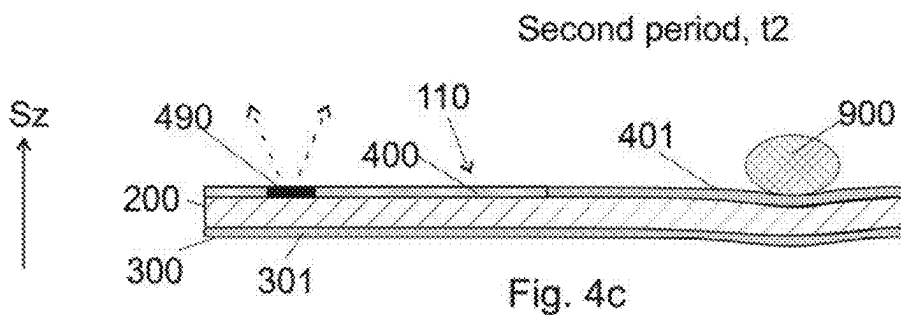
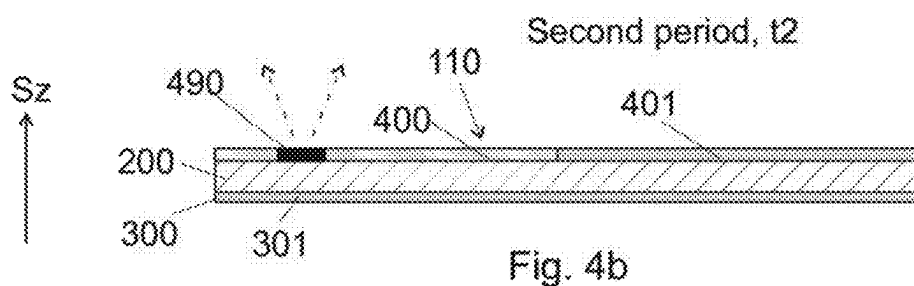
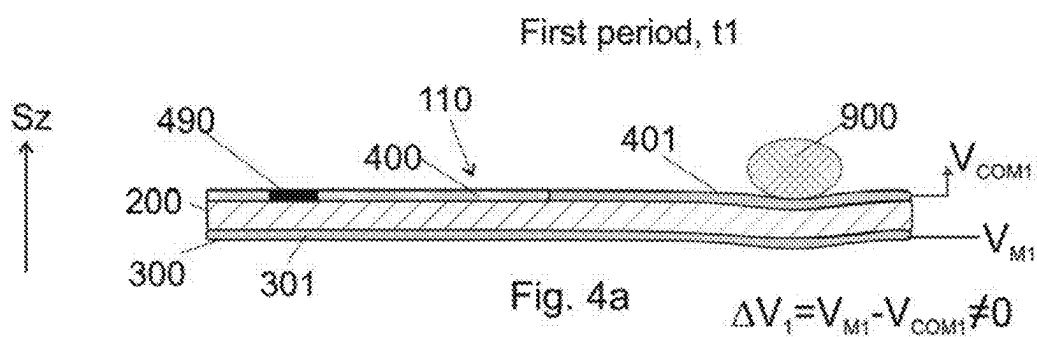
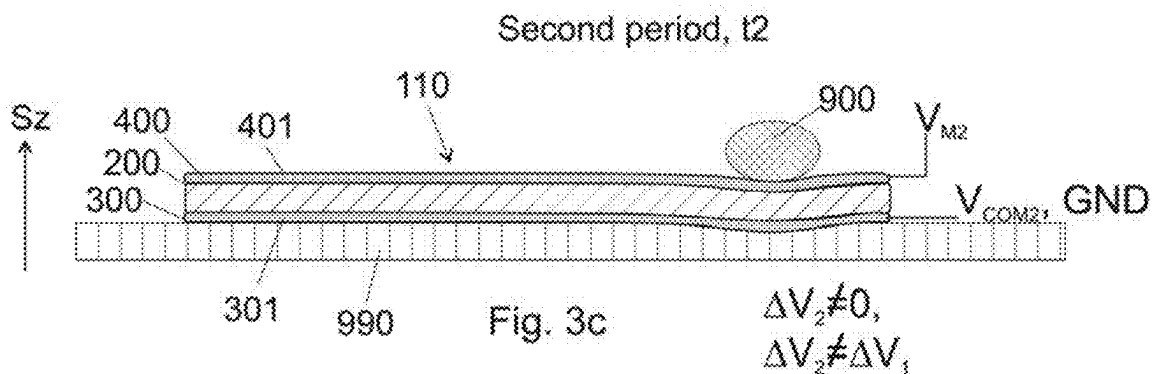
A method for operating a device has at least a first functionality and a second functionality, the first functionality being capacitive sensing of compression and the second functionality being another, different, functionality. For enabling the first functionality the device includes a first primary electrode, a first secondary electrode, and compressible and electrically insulating material between the first primary electrode and the first secondary electrode. During a first period of time, a capacitance of the first primary electrode relative to the first secondary electrode is measured for determining a compression of the compressible and electrically insulating material. The method includes detecting that the second functionality is enabled during a second period of time, or during a second period of time, enabling the second functionality, wherein the second functionality comprises another functionality than capacitive measurement of a quantity indicative of the compression. An arrangement performs the method.

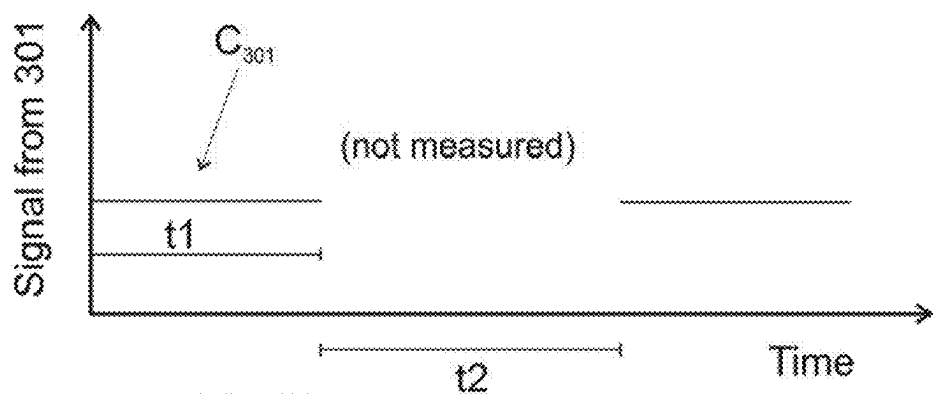
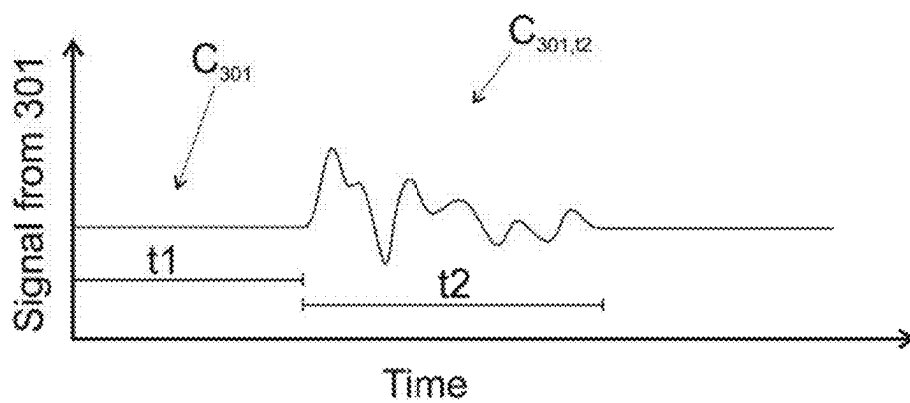
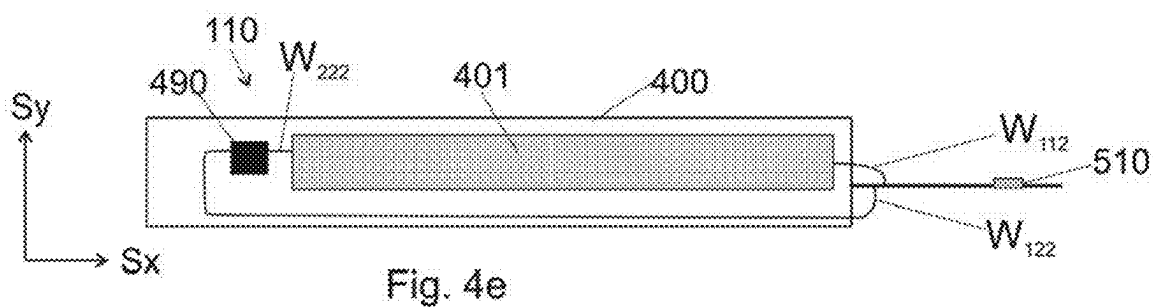
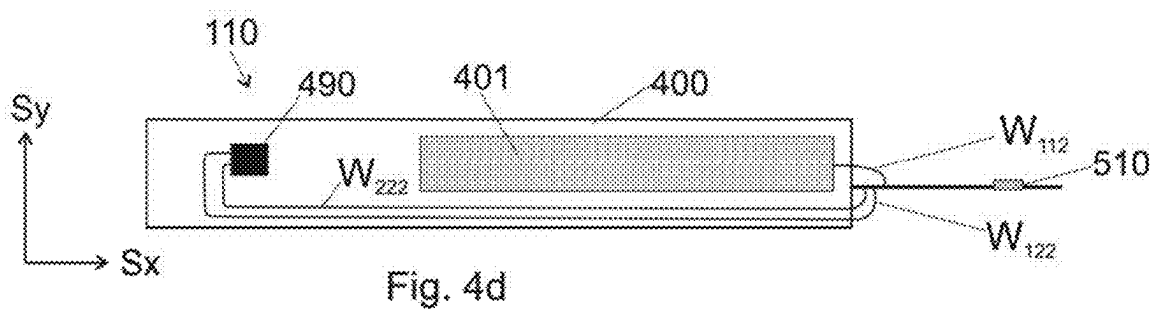












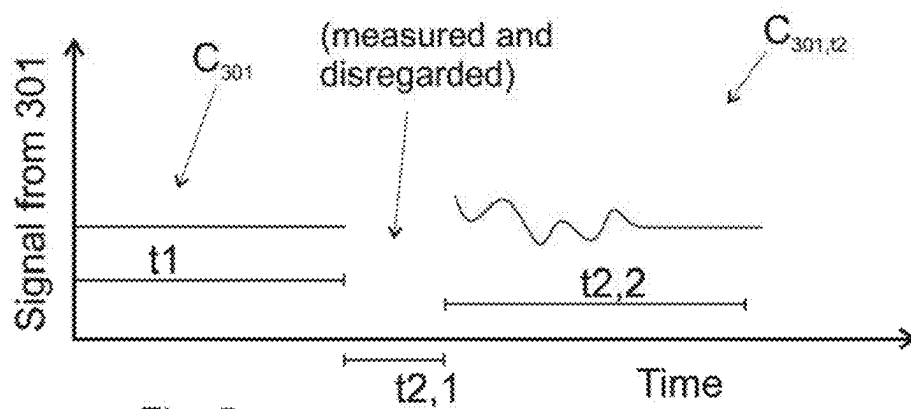


Fig. 5c

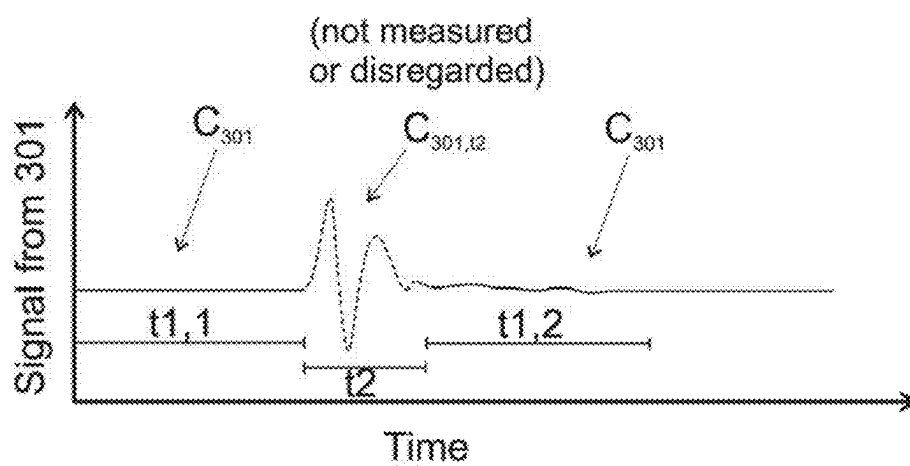
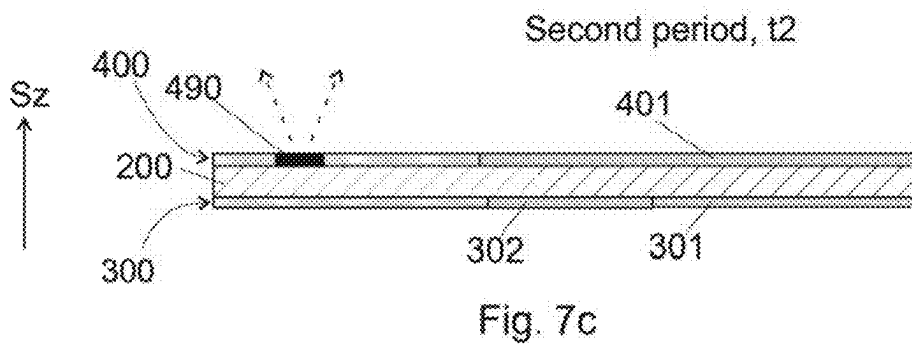
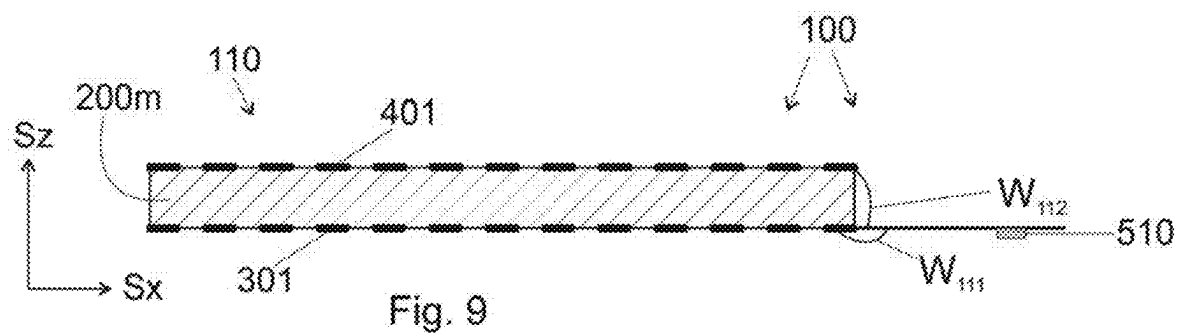
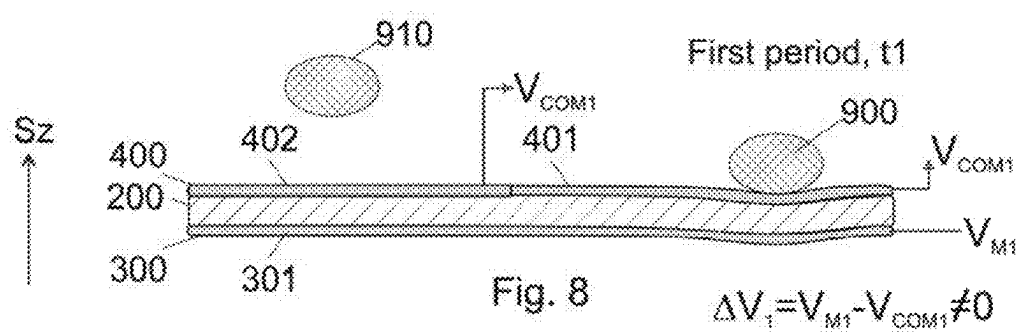


Fig. 5d





A METHOD FOR OPERATING A LAYERED DEVICE AND AN ARRANGEMENT COMPRISING A LAYERED DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a National Stage Application of PCT/FI2023/050237, filed May 2, 2023, which claims priority to Application No. 22171703.6, filed May 5, 2022 in Europe, and which applications are incorporated herein by reference. To the extent appropriate, a claim of priority is made to each of the above-disclosed applications.

TECHNICAL FIELD

[0002] The invention relates to a device that comprises a functionality of detecting compression capacitively. The compression is caused by a load, i.e. force or a pressure, whereby the invention relates to a capacitive force and/or pressure sensors.

BACKGROUND

[0003] A capacitive compression sensor **100** comprises a primary electrode layer **300** comprising at least a first primary electrode **301**, a secondary electrode layer **400** comprising at least a first secondary electrode **401**, and a compressible layer **200** comprising dielectric material between the primary electrode layer **300** and the secondary electrode layer **400**. An operating principle of such a sensor is well known e.g. from the document WO 2018/011464.

[0004] In short, the capacitance of the first primary electrode **301** relative to the first secondary electrode **401** depends on a distance between the electrodes, the distance being indicative of compression of the compressible layer **200** between the electrodes. The compression is indicative of a pressure and/or a force at that point by Hooke's law.

[0005] Nowadays, different functionalities of a system are enabled by using different components for different functionalities. E.g. a device having a force/pressure sensing functionality and a lightning functionality has one layered part for sensing the force/pressure and another part, e.g. further layer for illuminating. This makes the devices reasonably complex.

[0006] Furthermore, the capacitive measurement principle is reasonably sensitive to electrical fields of the environment. Therefore, in order to obtain reliable results, particularly when also other functionalities are present, efficient electromagnetic shielding is used to shield the capacitive part from other parts. As an example, reference is made to the shielding layers **600a** and **600b** of FIG. **5b** of WO 2018/011464. The shielding makes the devices even more complex.

[0007] Complex systems cost more to manufacture and tend to be less reliable than simpler systems. There is therefore a need for simple multifunctional devices capable of detecting load.

SUMMARY

[0008] It has been found that the complexity can be reduced by integrating the other functionality or the other functionalities to such a layer of the device that comprises an electrode that is used for measuring the compression or in between them. Thus, the device is capable of detecting compression capacitively and it also has another function-

ality, other than detecting compression capacitively. The other functionality may be a measurement of another parameter, operating an actuator, operating a light source, or operating a heat source. The component for the other functionality, which may, but need not, be used also for measuring the compression, is arranged in or on such a layer that comprises an electrode that is used for measuring the compression or within the compressible material (or material layer) in between the electrodes.

[0009] When measuring the capacitance to determine compression, a first secondary electrode **401** is connected to a common potential (e.g. ground potential). Then, the capacitance of a first primary electrode **301** is measured relative to (at least) the first secondary electrode **401**. In prior art, the layer comprising the first secondary **401** electrode has been used only for the purpose of measuring compression, i.e. forming a reference electrode for the first primary electrode **301**.

[0010] However, it has now been found that the compression (determinable from the capacitance) needs not be measured all the time. Moreover, at such a time interval that the compression is not measured, the secondary electrode layer that comprises the first secondary electrode **401** can be used for other purposes, i.e. to enable the other functionality of the device.

[0011] Moreover, shielding from external objects can be, at least to some extent, done by the electrode(s) **401**, **402** of the secondary electrode layer, when the device **110** is arranged so that the object that causes the load to the device is arranged on the side of the secondary electrode layer. Furthermore, shielding from the effects of the second functionality is not required, because the second functionality is taken into account in the determining of the compression of the compressible layer **200**. The second functionality may be taken into account by signal analysis or by controlling the times when the second functionality is active.

[0012] Thus, the invention relates, on one hand, to a method for operating a device **110** having at least a first functionality and a second functionality, the first functionality being capacitive sensing of compression and the second functionality being another, different, functionality. The method may be performed by a control unit **510**. The control unit may be, but need not be, part of the device **110**. The control unit **510** may be comprised by an arrangement **100** that comprises the device.

[0013] The invention is described in more specific terms in claims **1** and **10**.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. **1a** shows a layered device having at least two functionalities, one of which is measurement of compression, and an arrangement further comprising control unit,

[0015] FIG. **1b** shows a layered device having at least two functionalities, one of which is measurement of two compressions, and an arrangement further comprising a connector,

[0016] FIG. **1c** shows an arrangement comprising a layered device having at least two functionalities, one of which is measurement of two other capacitances, and an arrangement further comprising a control unit,

[0017] FIG. **1d** shows an arrangement comprising a layered device having at least two functionalities, one of which is measurement of compression of two layers,

[0018] FIG. 1e shows an arrangement comprising a layered device, the layers being coated,

[0019] FIG. 2a shows an arrangement comprising a layered device having at least two functionalities, one of which is measurement of compression, and the other can be enabled by using an activatable component,

[0020] FIG. 2b shows the arrangement of FIG. 2a, the device being coated,

[0021] FIG. 2c shows an arrangement comprising a layered device having two electrodes in the secondary electrode layer and an activatable component,

[0022] FIG. 2d shows, in a side view, an embodiment, wherein an activatable component also serve as an electrode,

[0023] FIG. 2e shows, in a top view, an embodiment, wherein an activatable component also serves as the first secondary electrode,

[0024] FIG. 3a shows a layered device having a first secondary electrode for the purpose of another functionality than sensing compression, the device sensing a compression using a first primary electrode,

[0025] FIG. 3b shows the layered device of FIG. 3a when the device is sensing another parameter than compression and the first secondary electrode is used for the other functionality,

[0026] FIG. 3c shows the layered device of FIG. 3a when the device is sensing another parameter than compression, even if the device is being compressed, and the first secondary electrode is used for the other functionality,

[0027] FIG. 4a shows, in a side view, a layered device having an activatable component for the purpose of another functionality than sensing compression, the device sensing a compression using a first primary electrode,

[0028] FIG. 4b shows the layered device of FIG. 4a when the activatable component is activated for the purpose of using it,

[0029] FIG. 4c shows the layered device of FIG. 4a when the activatable component is activated for the purpose of using it while the device is being compressed,

[0030] FIG. 4d shows, in a top view, a possible wiring of the secondary electrode layer of the device of FIG. 4a,

[0031] FIG. 4e shows, in a top view, a possible wiring of the secondary electrode layer of the device of FIG. 4a,

[0032] FIG. 5a shows a signal obtainable from the primary electrode during a first period t1, when the other functionality is disabled, and during a second period t2, when the other functionality is enabled,

[0033] FIG. 5b shows a signal obtainable from the primary electrode during a first period t1, when the other functionality is disabled, and not measuring a signal from the primary electrode during a second period t2, when the other functionality is enabled,

[0034] FIG. 5c shows a signal obtainable from the primary electrode during a first period t1, when the other functionality is disabled; not measuring a signal from the primary electrode during a beginning of the second period t2, when the other functionality is enabled; and measuring a signal from the primary electrode during an end of the second period t2, when the other functionality is enabled,

[0035] FIG. 5d shows a signal obtainable from the primary electrode during a first period t1, when the other functionality is disabled; first measuring a signal from the primary electrode during a second period t2; and measuring a signal from the primary electrode during a secondary first period, when the other functionality is enabled,

[0036] FIG. 6 shows connecting the device 110 to a first control unit 510 and to a second control unit 530,

[0037] FIGS. 7a to 7c show a device having an activatable component and two electrodes in a primary electrode layer for measuring compression from to different locations,

[0038] FIG. 8 shows a device having two electrodes in a secondary electrode layer and using them for purpose of shielding when measuring compression, and

[0039] FIG. 9 shows a device having at least two functionalities, one of which is measurement of compression, and an arrangement further comprising control unit.

[0040] In the figures, Sz denotes a direction of a thickness of a device 110. A direction Sx is perpendicular to Sz. A direction Sy is perpendicular to both Sz and Sx.

DETAILED DESCRIPTION

[0041] Most of the embodiments of the invention are hereinbelow explained using a layered device 110 as an example. However, a layered structure is not essential as disclosed later in connection with FIG. 9, which shows a device 110 and an arrangement 100.

[0042] FIG. 1a shows an arrangement 100 for operating a layered device 110. The layered device 110 has at least two different functionalities, i.e. at least a first functionality and a second functionality. The first functionality is capacitive sensing of compression and the second functionality is another, different, functionality. The compression is a result of a load, i.e. a pressure, which is distributed force, or a force, which is concentrated. Thus, the first functionality is also a capacitive measurement of at least load, i.e. force and/or compression.

[0043] To enable the first functionality, the device 110 comprises a primary electrode layer 300 comprising at least a first primary electrode 301. Referring to FIG. 1a, the primary electrode layer 300 may comprise only one electrode 301. However, referring to FIGS. 1b to 1d, 2c, and 7a to 7c, in an embodiment, the primary electrode layer 300 comprises the first primary electrode 301 and a second primary electrode 302. This has the effect that compression can be measured from the different locations defined by the different electrodes 301, 302 of the primary electrode layer 300. Naturally, the primary electrode layer 300 may comprise further electrodes as known from prior art

[0044] To enable the first functionality, the device 110 further comprises a secondary electrode layer 400 comprising at least a first secondary electrode 401. Referring to FIG. 1a, the secondary electrode layer 400 may comprise only one electrode 401. However, referring to FIGS. 1c, 1d, 2c, and 8, in an embodiment, the secondary electrode layer 400 comprises the first secondary electrode 401 and a second secondary electrode 402. This has the effect that the electrodes 401, 402 of the secondary electrode layer 400 define more locations on an area. Thus, if the secondary electrodes 401, 402 are used for the other functionality (e.g. sensing touch or pulse) this other quantity can be determined with better spatial accuracy, i.e. at the locations defined by the electrode 401, 402. Moreover, when using the first functionality, the electrodes 401, 402 can be used for shielding, as will be detailed below.

[0045] To enable the first functionality, the device 110 further comprises a compressible layer 200. The compressible layer 200 is arranged between the primary electrode layer 300 and the a secondary electrode layer 400. More specifically, at least a part of the compressible layer 200 is

arranged between the first primary electrode **301** and the first secondary electrode **401**. The compressible layer **200** comprises compressible and electrically insulating material **200m**. More specifically, at least some compressible and electrically insulating material **200m** is arranged between the first primary electrode **301** and the first secondary electrode **401**. An electrode (**301**, **401**) may be arranged in the layer **200** whereby the material **200m** may be arranged on both sides of the electrode (**301**, **401**).

[0046] The first functionality of the device **110** is based on the capacitive measurement principle of compression. As well known, a capacitance of a capacitor formed by two electrodes is related to their area, a dielectric constant of material in between them, and a mutual distance of the electrodes. The capacitance may depend also on other factors, e.g. a dielectric constant of material in the vicinity of the electrodes, elsewhere than between them. FIGS. **3a** and **4a** show an object **900** (e.g. a finger of a user) pressing the first secondary electrode **401** against the first primary electrode **301**. Thus, the material of the compressible layer **200** becomes compressed, whereby the distance between the electrodes **301**, **402** decreases, and the capacitance C_{301} of the first primary electrode **301** relative to the first secondary electrode **401** changes (i.e. increases). Even without any compression, the presence of the object **900** near one of the electrodes **401**, **301** may slightly affect the capacitance between them.

[0047] Referring to FIGS. **3a** and **4b**, the capacitance C_{301} of the first primary electrode **301** relative to the first secondary electrode **401** is measured during a first period of time **t1**. To measure the capacitance C_{301} , an embodiment of a method for operating the layered device **110** comprises arranging a first potential difference DV_1 between the first primary electrode **301** and the first secondary electrode **401**. The first potential difference DV_1 is non-zero. The first potential difference DV_1 is arranged by connecting the first secondary electrode **401** to a first common potential V_{COM1} , and conducting a first measurement potential V_{M1} to the first primary electrode **301**. The first measurement potential V_{M1} is different from the first common potential V_{COM1} . The first potential difference DV_1 is, by definition, a difference ($V_{M1} - V_{COM1}$) between the potential of the first primary electrode **301** and the potential of the first secondary electrode **401**. The embodiment further comprises measuring the capacitance C_{301} of the first primary electrode **301** relative to (at least) the first secondary electrode **401** for the purpose of determining a compression of the compressible layer **200**. As detailed above, the capacitance may depend on other factors, too. During the first period of time **t1**, the second functionality of the device **110** is disabled for not disturbing the capacitive measurements of compression. Preferable, the first common potential V_{COM1} is a ground potential GND. The first measurement potential V_{M1} may result from direct current (DC), whereby the sign of the first potential difference DV_1 can be constant throughout the first period of time **t1**, or the first measurement potential may be a result of alternating current (AC). Preferably, the sign of the first potential difference DV_1 is constant throughout the first period of time **t1**. It is noted, that if the capacitance C_{301} is measured by charging or discharging the capacitor, the potential of the capacitor changes during charging/discharging.

[0048] The first common potential V_{COM1} may have the effect of shielding the compressible layer **200** and thus also

shielding the effect of a foreign object **910** on the primary capacitance C_{301} affected primarily by the load caused by an object **900** (see FIG. **8**). To have the shielding effect, in an embodiment, the device **110** is used such that the object **900** that causes the load causing the compression of the compressible layer **200** is, during the first period of time **t1**, closer to the secondary electrode layer **400** than to the primary electrode layer **300** (see FIGS. **3a**, **4a**, and **8**). The object **900** may press the first secondary electrode **401** or some material **820** arranged on top of the electrode **401**. In an embodiment of the method, the device **110** is arranged on a support **990** so that the primary electrode layer **300** is closer to the support **990** than the secondary electrode layer **400** (see FIG. **3a**). Naturally, the compression of the compressible layer **200** is a result of both the support **990** and the object **900**. However, to have the shielding effect, in an embodiment, the device **110** is used such that the object **900** that causes the load causing the compression of the compressible layer **200** is configured to approach the device **110** from the side of the secondary electrode layer **400**. Thus, in an embodiment of the method, during the first period of time **t1**, at least a part of the device is arranged between a support **990** and an object **900** causing the compression of the compressible layer **200** such that the primary electrode layer **300** faces the support **990** and the secondary electrode layer **400** faces the object **900** (see FIG. **3a**). Moreover, during the second period of time, the primary electrode layer **300** faces the support **990**. Moreover, electrodes **401**, **402** of the secondary electrode layer **400** may be used for shielding during the first period of time **t1**, as detailed below.

[0049] Correspondingly, an arrangement **100** comprises a device **110** configured to be used such that the object **900** causing the compression of the compressible layer **200** is configured to approach the device **110** from the side of the secondary electrode layer **400**.

[0050] These electric potentials can be generated by a control unit **510** (see e.g. FIG. **1a**). In the alternative, the electric potentials can be guided to the electrodes through a connector **520** (see FIG. **1b**). Thus, a method for operating the layered device **110** needs not be carried out by the arrangement **100** of FIG. **1a**.

[0051] In an embodiment, the arrangement **100** comprises the control unit **510**. The control unit **510** is, during the first period of time **t1**, configured to arrange the first potential difference DV_1 between the first primary electrode **301** and the first secondary electrode **401** by connecting the first secondary electrode **401** to a first common potential V_{COM1} and conducting a first measurement potential V_{M1} to the first primary electrode **301**, the first measurement potential V_{M1} being different from the first common potential V_{COM1} . The control unit **510** is, during the first period of time **t1**, configured to measure a capacitance C_{301} of the first primary electrode **301** relative to (at least) the first secondary electrode **401** for determining a compression of the compressible layer **200**.

[0052] Moreover, the device **110** comprises at least a primary first primary wire W_{111} that is configured to guide the first measurement potential V_{M1} to the first primary electrode **301**. Moreover, the device **110** comprises at least a primary first secondary wire W_{112} that is configured to guide the first common potential V_{COM1} to the first secondary electrode **401**. Reference is made to FIG. **1a**.

[0053] As for the numbering of the wires, a wire W_{ijk} is arranged in the layer **k** (**1** for primary electrode layer **300** and

2 for secondary electrode layer 400), connected to the component/electrode having the index j of that layer, the wire being the i :th wire connected to that component/electrode.

[0054] While in general, only one wire needs to be guided to an electrode 301, 401, it is noted already at this point, that the first primary electrode 301 or the first secondary electrode 401 may be used as an activatable component 490, such as a heater. FIGS. 2d and 2e show such an embodiment in a side view and top view, respectively, wherein the component is arranged in the secondary electrode layer 400. To activate the heating functionality, two wires are needed: one to guide a current to the component 490 and another to guide a current from the component 490. Clearly, the primary first secondary wire W_{112} may be used to guide current to the first secondary electrode 401. Thus, in an embodiment, where the first secondary electrode 401 is used as an activatable component 490, the device 110 comprises at least a secondary first secondary wire W_{212} that is configured to guide a current from the first secondary electrode 401, see FIG. 2e.

[0055] Even without two wires, an electrode (301, 401) may be used, in addition to sensing compression, as an antenna.

[0056] In general, an electrode, in particular the electrode 301 or 401, comprises electrically conducting material. An electrically conducting material has a resistivity of at most 1 m at a temperature of 23° C., preferably at most 0.1 m at a temperature of 23° C. Also the wires W_{111} , W_{112} , W_{212} , W_{122} , W_{222} comprise electrically conducting material in this sense.

[0057] As indicated above, the compressible layer 200 comprises material that is compressible and electrically insulating. As for the electrical insulation, a resistivity of the material of the compressible layer 200 is, in an embodiment, at least 10 m at a temperature of 23° C. This allows for not forming a short-circuit between the electrodes 301, 401. As for the compressibility, a Young's modulus of the material of the compressible layer is, in an embodiment, at most 3 GPa, preferably at most 1 GPa, such as from 0.05 MPa to 15 MPa, such as from 0.2 MPa to 5 MPa. These values are given at the temperature 23° C. This allows for sufficient compression also when the compression is generated by a human without tools (e.g. by pressing the device 110 with fingers, by standing on the device 110, or by laying or sitting on the device 110). Moreover, a compressible material can be compressed by at least 10% in a reversible manner. The reversibility of the compression is spontaneous, i.e. elastic.

[0058] Preferably, the device 110, including the electrode layers 300, 400 and the compressible layer 200, is stretchable. A stretchable material can be stretched by at least 10% in a reversible manner at a temperature of 20° C. The stretchable material can be stretched in such a way in a direction that is perpendicular to a direction of a thickness of the material. The reversibility of the stretching is spontaneous, i.e. elastic. This applies to the device 110, too. However, not all the parts of the device 110 need to be stretchable. E.g. the component 490 may be rigid. Even if the device 110 is stretchable such that it can be stretched by at least 10% in a reversible manner at a temperature of 20° C., it may comprise a rigid (non-stretchable) component.

[0059] More preferably, the device 110, including the electrode layers 300, 400 and the compressible layer 200, is conformable. Herein the term conformable refers to material

that is flexible, compressible, and stretchable. A planar flexible material can be bent to a radius of curvature of 10 mm without breaking the material at a temperature of 20° C. Moreover, the flexible material can be thereafter turned back to the planar form at a temperature of 20° C. without breaking the material. Thus, a planar conformable device 110 is flexible as indicated above, compressible in the direction S_z of its thickness as detailed above, and stretchable in a direction of the plane of the planar device 110 (i.e. in the direction(s) S_x and/or S_y). A planar conformable device 110 can be arranged to conform a surface of a sphere or a hemisphere having a radius of 10 cm at a temperature of 20° C. without introducing plastic (i.e. irreversible) deformations to the material. Naturally, the device 110 may be larger. Thus, at least a part of a conformable device may be arranged to conform a surface of a hemisphere having a radius of 10 cm at a temperature of 20° C. without introducing plastic (i.e. irreversible) deformations to the material.

[0060] The stretchability, preferably conformability, of the device 110 allows for its use in various applications. A stretchable or conformable device 110 can be arranged on many surfaces for detecting the compression (i.e. load in the meaning of force and/or pressure) and furthermore, the device has also the second functionality. Moreover, the device 110 may have a very simple structure. Examples of devices wherein the device 110 can be used include remote controllers, steering wheels, handles, handle bars, joysticks, dashboards (e.g. of vehicles), seats and tables.

[0061] The second functionality of the device 110 is enabled during a second period of time t_2 . The second functionality is enabled by using the first primary electrode 301, the first secondary electrode 401, a component 490 arranged in or on the primary electrode layer 300, or a component 490 arranged in or on the secondary electrode layer 400. The component 490 may be arranged in the compressible layer 200; or a part of the component 490 may be arranged in the compressible material 200m in between the electrodes 301, 401. In an embodiment, the second functionality is actively enabled by the same entity that performs the method. In an embodiment, the fact of the second functionality being enabled is detected by signal analysis. Thus, an embodiment comprises detecting that the second functionality is enabled.

[0062] The component 490 may be a rigid component (e.g. a microchip, a resistor, a transistor, optionally being arranged in a package such as a plastic package). The component 490 may be a flexible component on a film. The film may be attached to the primary electrode layer 300 or the secondary electrode layer 400. The component 490 may be a flexible component manufactured directly on the primary electrode layer 300 or the secondary electrode layer 400.

[0063] Preferably, the second functionality is enabled by using the first secondary electrode 401 or a component arranged in or on the secondary electrode layer 400; and the device 110 is used such that the object 900 that causes the load causing the compression of the compressible layer 200 is configured to approach the device 110 from the side of the secondary electrode layer 400 (see above). This has two effects. First, some types of second functionalities, e.g. capacitive measurements, cannot be used, if the primary electrode layer 300 would be arranged between the first secondary electrode 401 and the object 900 causing the change to the capacitance of the first secondary electrode

401, because the primary electrode layer would shield the secondary electrode layer from disturbance in the electric field. Therefore, capacitive measurements are preferably used for the second functionality only when the secondary electrode layer **400** is arranged between the primary electrode layer **300** and the object **900** causing the change to the capacitance of the first secondary electrode **401** (see FIG. 3*b* and above). Second, it is noted that the secondary electrode layer **400** does not prevent using an activatable component **490** in the primary electrode layer **300** (see FIG. 4*b*). For example, a heater (e.g. the first primary electrode **301**) arranged in the primary electrode layer **300** could be used also in the embodiment of FIG. 4*b*, and the heat generated in the primary electrode layer **300** would be, eventually, conducted all over the device **110**. However, typically the activatable component **490** is needed to form a signal or a response to a user interface, and the user interface is closer to the secondary electrode layer **400** than to the primary electrode layer **300**. For example, the secondary electrode layer **400**, or a material layer **820** (see FIG. 1*e*) arranged on top of the secondary electrode layer **400**, may form the user interface. Herein the term “on top” means that the secondary electrode layer **400** is arranged between the material layer **820** and the compressible layer **200** (see FIG. 1*e*). Therefore, to obtain a more prompt response on the user interface, preferably, the component **490** is arranged in or on the secondary electrode layer **400**.

[0064] In an embodiment, the second functionality comprises another function than measuring the same signal from which the compression is detected and in the same way as the compression is detected. More specifically, the second functionality comprises another functionality than capacitive measurement of a quantity such that the first secondary electrode **401** is connected to the first common potential V_{COM1} , and the first measurement potential V_{M1} is conducted to the first primary electrode **301**. Thus, in an embodiment, the second functionality is not a measurement using the same signal from which the capacitance C_{301} is determined during the first period of time.

[0065] As for a type of the second functionality of the device **110**, there are at least three options.

[0066] As a first option, the first secondary electrode **401** may be used to measure a capacitance of the first secondary electrode **401** relative to its surroundings including the first primary electrode **301**. This is shown in FIG. 3*b*. For example, the first secondary electrode **401** may be arranged on top of the device **110** so that an object **900** (e.g. a hand or a finger of a user) can be arranged close to the first secondary electrode **401**. When measuring the capacitance C_{401} of the first secondary electrode **401**, a second potential difference DV_2 is applied between the first secondary electrode **401** and the first primary electrode **301**. To enable measurement of the capacitance C_{401} , the second potential difference DV_2 is different from zero (i.e. not zero). Moreover, to ensure that the signal caused by the object **900** is readable from the capacitance C_{401} , the second potential difference DV_2 is different from the first potential difference DV_1 . Therefore, in this embodiment, the second functionality comprises another functionality than capacitive measurement of a quantity such that the first secondary electrode **401** is connected to the first common potential V_{COM1} , and the first measurement potential V_{M1} is conducted to the first primary electrode **301**.

[0067] More preferably, the second potential difference DV_2 is arranged, during the second period of time **t2**, so that the first primary electrode **301** is electrically connected to a second common potential V_{COM2} , which even more preferably is a ground potential GND. This has the beneficial effect that commonly also the object **900** is in an at least loose connection to a ground potential, whereby connecting the ground potential also to the first primary electrode **301** ensures that the effect cause by the object **900** is seen in the capacitance C_{401} .

[0068] As a second option, an activatable component **490** may be arranged in or on the secondary electrode layer **400** (or in the compressible layer **200**). The term activatable component refers to a transducer. The term activatable component may refer to an electric component that comprises only one lead to which a signal is applied or generated, such as an antenna. The antenna transduces an electric signal to an RF signal or vice versa. In such a case the activation involves feeding an electric current to the component **490** or receiving an electric current from the component **490**.

[0069] The term activatable component may refer to an electric component that comprises two leads between which a signal is applied or generated. For example, one lead may be for letting in current and another for letting out current. The component **490** may be activated by driving a current through these leads or forming a potential difference between the leads. In the alternative, the activatable component **490** may be activated by another form of energy that causes a potential difference to the leads which can be sensed as such or as an electric current. FIG. 4*b* shows an embodiment, wherein, during the second period of time **t2**, the component **490** is activated. In FIG. 4*b*, the arrows indicate illumination, which is caused by an electric current passing the component **490**. Examples of the component **490** will follow. Also in this embodiment, the second functionality comprises another functionality than capacitive measurement of a quantity such that the first secondary electrode **401** is connected to the first common potential V_{COM1} , and the first measurement potential V_{M1} is conducted to the first primary electrode **301**. However, as discussed below, when the second functionality is active, the method may also comprise capacitive measurement of a quantity such that the first secondary electrode **401** is connected to the first common potential V_{COM1} , and the first measurement potential V_{M1} is conducted to the first primary electrode **301**; but the second functionality comprises also at least something else. For example, the second functionality comprises something else than signal analysis of a signal measured from the electrodes (**301**, **401**) such that the first secondary electrode **401** is connected to the first common potential V_{COM1} , and the first measurement potential V_{M1} is conducted to the first primary electrode **301**.

[0070] As a third option, as indicated already above, a first secondary electrode **401** may serve as an activatable component **490** so that during the first period of time **t1**, the first secondary electrode **401** is electrically coupled only to the first common potential V_{COM1} (e.g. using one or both of the wires W_{212} , W_{112}) and during a second period of time **t2**, an electric current is driven through the first secondary electrode **401** using the wires W_{112} , W_{212} for activating the component **490**. Reference is made to FIG. 2*e*. For example, a heater may serve for the purposes of an electrode **401** and

an activatable component 490, which when applying an electric current, transduces the electric current to heat.

[0071] Concerning wiring of the components/electrodes, the first secondary electrode 401 need not serve for the purposes of supplying an electric current to the component 490 or receiving a current therefrom; however, it may. Referring to FIG. 4d, the primary first secondary wire W_{112} can be used to guide the first common potential V_{COM1} to the first primary electrode 401; and the primary second secondary wire W_{122} and the secondary second secondary wire W_{222} can be used to drive the component 490 to activate it or to receive an electric signal from the component upon activating it by another (non-electric) signal.

[0072] However, the first secondary electrode 401 may serve for the purposes of supplying an electric current to the component 490 or receiving a current therefrom. Referring to FIG. 4e, during the first period of time t1, the primary first secondary wire W_{112} can be used to guide the first common potential V_{COM1} to the first secondary electrode 401. Moreover, the primary second secondary wire W_{122} is connected to the first lead of the component 490, while the secondary second secondary wire W_{222} is connected to a second lead of the component 490 and to the first secondary electrode 401. Thus, during the second period of time t2, the secondary second secondary wire W_{222} , the first secondary electrode 401, and the primary first secondary wire W_{112} , in combination, form a conductor from the component 490 to the control unit 510 or the connector 520. Thus, the wires W_{122} and W_{222} can be used to drive the component 490 to activate it or to receive an electric signal from the component 490 upon activating it by another (non-electric) signal. The wires W_{112} and W_{122} are connected to the controller 510 or the connector 520 for receiving or feeding the electric signal from/to the component 490.

[0073] Therefore, an embodiment of a method for operating the layered device 110 comprises during a second period of time t2 [A] detecting that the second functionality is enabled, [B] enabling the second functionality of the device 110 by arranging a second potential difference DV_2 between the first primary electrode 301 and the first secondary electrode 401, the second potential difference DV_2 being different from the first potential difference DV_1 and different from zero, [C] activating a component 490 of the layered device 110, and/or [D] arranging an object to a galvanic contact with the first secondary electrode 401. As indicated above, preferably, the first secondary electrode and/or the component 490 is/are arranged in or on the secondary electrode layer 400. Moreover, concerning [C], an antenna may be activated. However, preferably, the activating of the component 490 comprises feeding an electric current to the component 490 through a wire and simultaneously receiving an electric current from the component 490 through another wire.

[0074] Concerning the option [A], this may be used e.g. when the component 490 is a transducer that transduces a non-electric signal to an electric signal, or if the component 490 is another transducer that is controlled by a different entity than the one performing the method (e.g. another control unit 530 than the control unit 510 performing the method). Concerning the option [B], this may be used e.g. when the second functionality is a capacitive measurement by the first secondary electrode 401. Concerning the option [C], this may be used e.g. when the second functionality involves activating a component 490 and the component can

be activated by the same entity (e.g. the control unit 510) that performs the method. Concerning the option [D], this may be used e.g. when the second functionality is an electric measurement by the first secondary electrode 401.

[0075] Moreover, as indicated in the background, the capacitive measurement of compression is reasonably sensitive. Therefore, the enabling of the second functionality would disturb the measurement of the capacitance C_{301} , if the capacitance was measured also during the second period of time t2. The capacitance C_{301} may be measured only during the first period of time t1, whereby the enabling of the second functionality only during the second period of time will not disturb these measurements. However, the capacitance C_{301} may be measured also during the second period of time, when the second functionality is enabled. In any case, the embodiment of the method comprises taking into account the enabling of the second functionality in the determining or the compression of the compressible layer 200. The taking into account the enabling of the second functionality in the determining or the compression of the compressible layer 200 may include measuring the capacitance C_{301} also during the second period of time t2, and correcting the measured capacitance or determining the compression therefrom with the knowledge that the second functionality has been enabled. If, for example, the enabling of the second functionality only increases the capacitance by a substantially constant value, this can be taken into account when determining the compression from the capacitance. As another example, in case the enabling of the second functionality shows as a fluctuating signal in the capacitance, such fluctuation may be filtered to obtain a corrected value of the capacitance (or the compression directly).

[0076] Concerning the arrangement 100, in an embodiment, the control unit 510 is configured to, during a second period of time t2, enable the second functionality of the device 100 by arranging a second potential difference DV_2 between the first primary electrode 301 and the first secondary electrode 401, the second potential difference DV_2 being different from the first potential difference DV_1 and different from zero and/or by activating a component 490 arranged in or on the secondary electrode layer 400. In addition, the control unit 510 is configured to take into account the enabling of the second functionality in the determining the compression of the compressible layer 200.

[0077] Referring to FIG. 6, the layered device 110 may be operated simultaneously by a first control unit 510 and a second control unit 530. Both the control units 510, 530 may be connected to the device through the connector 520. In the alternative, the layered device 110 may be operated by only one control unit 510 (see FIG. 1a).

[0078] From the point of view of the first control unit 510, how the enabling of the second functionality is taken into account depends on whether the first control unit can control the component 490; or is it a responsibility of another, independent, entity to operate the component 490. For example, the second control unit 530 may operate the component 490 independent of the first control unit. In the alternative, a light source (e.g. the Sun) may operate independent of the first control unit 510, and the component 490 may be a photovoltaic cell.

[0079] In case the first control unit 510 can control the enabling (i.e. activating) and deactivating the second functionality of the device 110, preferably when changing from the first period t1 to the second period t2, the second

functionality is controlled to be activated so that it remains activated during the second period of time t_2 . Moreover, because the second functionality can be controlled, the control unit **510** needs not measure the capacitance C_{301} (and the compression) during the second period of time t_2 . Thus, referring to FIG. **5b**, in an embodiment, the second functionality is controlled to be deactivated during the first period of time t_1 , and the capacitance C_{301} is measured at that period of time t_1 . Moreover, in the embodiment, the second functionality is controlled to be activated during the second period of time t_2 , and the capacitance C_{301} is not measured at that period of time t_2 or if measured, the value thereof is disregarded or corrected. Thus, an embodiment comprises controlling the second functionality such that during the first period of time the second functionality is deactivated and during the second period of time the second functionality is activated.

[0080] During the second period of time t_2 , the capacitance C_{301} may be both measured and not measured. Referring to FIG. **5c** during the first period of time t_1 , the capacitance C_{301} can be measured as detailed in connection with FIG. **5a**. Thereafter, during a primary part $t_{2,1}$ of the second period of time, the second functionality may be active, and the capacitance not measured. Moreover, thereafter, during a secondary part $t_{2,2}$ of the second period of time, the second functionality may be active, and the capacitance C_{301} may be also measured. The enabling of the second functionality may be taken into account by correcting the signal and/or in the way that the compression is determined from the signal. The first period of time t_1 of FIG. **5a** may correspond to the measurement as depicted in FIG. **3a**. The primary part $t_{2,1}$ and the secondary part $t_{2,2}$ of the second period of time t_2 may correspond to the measurement as depicted in FIG. **3c**.

[0081] In practice, it may be that the disturbances are most prominent at the instance of time, when the other functionality is turned on, and the disturbances decrease with time, even if the other functionality remains enabled, as shown in FIG. **5a**. Thus, in accordance with FIG. **5c** and what has been said above, the measurement results may be disregarded (or no measurements of the capacitance C_{301} be made) only for a short period of time (i.e. during the primary part $t_{2,1}$ of the second period of time) immediately after the other functionality is turned on. However after a short while, the measurement results concerning the capacitance C_{301} may be taken into account, because the disturbance is not necessarily significant; as readable from FIG. **5c**.

[0082] To clarify, FIG. **5d** shows an embodiment, wherein the capacitance C_{301} is measured during a primary first period of time $t_{1,1}$, when the second functionality is disabled. Thereafter, the second functionality is enabled, which would show as a disturbance in the signal, as shown by the dotted line at the second period of time t_2 . However, the disturbance may be mainly related to the onset of switching on the second functionality. Correspondingly reasonably shortly after switching on the second functionality, it does not significantly affect the measurements. Therefore, during a secondary first period of time $t_{1,2}$, even if the second functionality is enabled, the capacitance C_{301} may be measured and used even without taking into account the enabling of the second functionality. However, as detailed above, during the second period of time (i.e. at the time of switching on the second functionality), the effect of the second functionality is taken into account.

[0083] Thus, in an embodiment, the control unit **510** is configured to control the second functionality such that during at least a part of the first period of time t_1 the second functionality is deactivated and during the second period of time t_2 the second functionality is activated.

[0084] However, referring to FIG. **5a**, in an embodiment, first control unit **510** cannot control the activating and deactivating of the second functionality of the device **110**. Instead, the second functionality may be activated and deactivated by another control unit **530** or another entity. Thus, the first control unit **510** (or another entity practicing the method) only knows the status of the second functionality from a signal measured from the first primary electrode **301** or from a data message sent between the control units **530** and **510**, to inform the other about the (upcoming/done) state change. The embodiment comprises measuring a signal from the first primary electrode **301** also during the second period of time t_2 , and determining, from the signal, that the second functionality has been enabled. The signal may be the capacitance C_{301} . Thus, an embodiment comprises measuring a capacitance C_{301,t_2} of the first primary electrode **301** relative to the first secondary electrode **401** also during the second period of time t_2 , and determining, from the capacitance C_{301,t_2} , that the second functionality has been enabled. Moreover, after determining that the second functionality has been enabled, the embodiment comprises compensating from signal, such as the capacitance C_{301,t_2} , the effect of the second functionality or disregarding the signal. Referring to FIG. **5a**, during the second period of time t_2 , the activity of the second functionality generates disturbance to the signal, which needs to be taken into account, or the signal during the second period t_2 may be disregarded. It is noted that even if the capacitance C_{301} may be measured during the second period of time t_2 , the second functionality includes at least some other functionality. Moreover, even if the method may comprise compensating from signal, such as the capacitance C_{301,t_2} , the effect of the second functionality, the second functionality comprises something else than signal analysis of a signal measured from the electrodes (**301**, **401**) such that the first secondary electrode **401** is connected to the first common potential V_{COM1} , and the first measurement potential V_{M1} is conducted to the first primary electrode **301**; namely the thing that disturbs the signal.

[0085] Therefore, in an embodiment, the control unit **510** is configured to measure a signal (e.g. the capacitance C_{301}) from the first primary electrode during the second period of time and to determine, from the signal, that the second functionality has been enabled. Finally, the control unit is configured to compensate for the effect of the second functionality on the capacitance of the first primary electrode or disregard the capacitance measured during the second period of time or both. Referring to FIG. **5c**, at a primary second period of time $t_{2,1}$ the capacitance may be disregarded and at a secondary second period of time $t_{2,2}$ the effect of the second functionality on the capacitance can be compensated for.

[0086] Moreover, it is possible that the first control unit **510** gets information about the activity of the second functionality otherwise. E.g. if activating the second functionality involves raising a potential at a lead of a component **490**, the potential of that lead may be guided directly as an input signal to the first control unit **510**. Thus, from this input signal, the first control unit **510** can determine that second functionality is enabled. Correspondingly, the capacitance

C_{301} (or another signal) would not need to be measured during the second period of time. Or, if measured, the capacitance could be disregarded. Reference is made to FIG. 5b. Evidently, the input signal would be detectable from the component 490 arranged in or on the primary or secondary electrode layer (300, 400) or the electrode (301, 401). Thus, an embodiment comprises measuring a signal from the primary electrode layer 300 or from the secondary electrode layer 400 during the second period of time t2; determining, from the signal, that the second functionality has been enabled; and one of [i] using the signal to compensate for the effect of the second functionality of the capacitance of the first primary electrode $C_{301,r2}$ or [ii] disregarding the capacitance $C_{301,r2}$ measured during the second period of time t2.

[0087] As detailed above, preferably, the input signal would be detectable from the component 490 arranged in or on the secondary electrode layer 400 or the first secondary electrode 401. Thus, an embodiment comprises measuring a signal from the secondary electrode layer 400 during the second period of time t2; determining, from the signal, that the second functionality has been enabled; and one of [i] using the signal to compensate for the effect of the second functionality of the capacitance of the first primary electrode $C_{301,r2}$ or [ii] disregarding the capacitance $C_{301,r2}$ measured during the second period of time t2.

[0088] Therefore, in an embodiment, the control unit 510 is configured to measure a signal (e.g. a potential of a lead of a component 490) from the secondary electrode layer 400 during the second period of time t2 and to determine, from the signal, that the second functionality has been enabled. Finally, the control unit is configured to compensate for the effect of the second functionality on the capacitance of the first primary electrode or to disregard the capacitance measured during the second period of time.

[0089] Preferably, the inactivity of the second functionality during the first period of time is also determined. The inactivity may be measured from the capacitance C_{301} , which is measured during the first period of time t1. Thus, an embodiment comprises determining, from the capacitance C_{301} measured during the first period of time, that the second functionality has been disabled during the first period of time. The inactivity may be measured from input signal readable from the secondary electrode layer (see above). Thus, an embodiment comprises measuring a signal from the secondary electrode layer during the first period of time and determining, from the signal, that the second functionality has been disabled.

[0090] Correspondingly, in an embodiment, preferably, the control unit 510 is configured to determine, from the capacitance measured during the first period of time, that the second functionality has been disabled during at least a part of the first period of time and/or measure a signal from the secondary electrode layer during the first period of time and determining, from the signal, that the second functionality has been disabled during at least a part of the first period of time.

[0091] Concerning the second functionality, irrespective of whether it becomes enabled spontaneously or is activated e.g. by the control unit 510, in an embodiment, the second functionality involves transducing an electric signal (current or voltage) to another signal (e.g. optical signal, heat, or mechanical signal) or transducing a non-electric signal to an electric signal. In this embodiment, the device 110 comprises an activatable component 490, which may be the first

secondary electrode 401, that is arranged in or on the secondary electrode layer 400. The device 110 comprises a primary wire (W_{112} , W_{122}) electrically connected to the component 490. If the component 490 is an antenna, only one wire suffices, and the component may transduce an RF-signal to electric signal and/or vice versa.

[0092] Preferably, two electrically conducting wires extend from the component 490. Referring to FIGS. 2e, 4d, and 4e, in an embodiment the device 110 comprises a primary wire (W_{112} , W_{122}) electrically connected to the component 490 and a secondary wire (W_{212} , W_{222}) electrically connected to the component 490. In an embodiment, the component 490 is configured to transduce an electric signal to another signal, the primary wire (W_{112} , W_{122}) is configured to feed an electric current to the component 490, and the secondary wire (W_{212} , W_{222}) is configured to let out the electric current from the component 490. In an embodiment, the component 490 is configured to transduce a non-electric signal to an electric signal, and the electric signal is detectable from the primary wire (W_{112} , W_{122}) and the secondary wire (W_{212} , W_{222}). As indicated above, the primary wire W_{1jk} is connected to the component j of the layer k; and the secondary wire W_{2jk} is connected to the component j of the layer k.

[0093] Concerning the arrangement 110 comprising the control unit 510, therein the primary wire (W_{112} , W_{122}) is connected to the control unit 510. When the device comprises the secondary wire (W_{212} , W_{222}), the secondary wire (W_{212} , W_{222}) is connected to the control unit 510 (optionally via another wire and an electrode). However the method may be performed by another control unit attachable to the connector 520. Correspondingly, the primary and secondary wires may be connected to the connector 520.

[0094] In an embodiment of the method the second functionality involves transducing, by using the component 490, (i) an electric current or voltage to another, i.e. non-electric, form of a signal or (ii) a non-electric signal to an electric current or voltage.

[0095] Examples of the activatable component 490 that transduce another form of energy than electricity to an electric signal, and need at least one lead for functioning, include (i) galvanic contact pads (e.g. for driving electricity to an object) and (ii) antennas. An electrode (401, 301) may serve as a galvanic contact pad. In such an embodiment, at least a part of at least one of the component 490, the first primary electrode 301, and the first secondary electrode 401 is exposed. Because of the exposure, an object can be arranged in galvanic contact with the component 490, the first primary electrode 301, and/or the first secondary electrode 401. The galvanic contact can be formed without removing solid material of the device 110 or penetrating through solid material of the device 110. This applies to the device 110, the arrangement 100, and the method. It is possible that some measurements require presence of two exposed electrodes, and an electric signal is readable between these two electrodes once parts of an object are arranged in galvanic contact with both these electrodes. E.g. some heart-rate monitor operate according to this principle. In the alternative, an arrangement of two exposed electrodes (each having only one lead) can be seen as a (single) component having two leads.

[0096] Examples of the activatable component 490 that transduce another form of energy than electricity to an electric signal, and need at least two leads for functioning,

include (i) piezoelectric components, e.g. sensors, that, when compressed, induce electric potential difference; (ii) photodetectors, including photovoltaic cells, that, when receiving light, induce electric current, (iii) thermoelectric components that, when imposed to a temperature gradient, induce electric current, (iv) microphones that, when imposed mechanical vibration, induce electric current.

[0097] Preferably, the second functionality involves feeding an electric current to the component 490 and receiving an electric current from the component 490.

[0098] A transducer, which is activated by feeding current to it may be e.g. a heater, an antenna, a light emitter, a display, or a sensor, such as a temperature sensor, a vital signs sensor, an inductive sensor, a piezoresistive sensor, or a strain gauge. The vital signals sensor may be e.g. a galvanic contact pad (or an arrangement with at least two contact pads) configured to form an electric contact (or electric contacts) with a skin of a user.

[0099] A transducer, which is activated by feeding current to it and receiving current therefrom may be e.g. a heater, a light emitter, a display, or a sensor, such as a temperature sensor, a vital signs sensor, an inductive sensor, a piezoresistive sensor, or a strain gauge.

[0100] As opposed to an activatable component comprising two leads, referring to FIGS. 3a and 3b, the second functionality can be achieved by using the first secondary electrode 401. FIG. 3a shows the first functionality in this case, i.e. measurement of compression caused by the object 900. These measurements are performed as discussed above.

[0101] The second functionality can also be enabled by using the first secondary electrode 401 (or the first primary electrode 301) as an antenna. Or conversely, some kind of antenna may be used as an electrode.

[0102] FIG. 3b shows another type of the second functionality using the first secondary electrode 401. In this case, the second functionality is enabled by arranging a second potential difference DV_2 between the first primary electrode 301 and the first secondary electrode 401. The second potential difference DV_2 is different from the first potential difference DV_1 and different from zero. The embodiment further comprises measuring a secondary capacitance C_{401} of the first secondary electrode 401.

[0103] The second potential difference DV_2 not being zero ensures that the secondary capacitance C_{401} can be measured. The second potential difference DV_2 not being the same as the first potential difference DV_1 ensures that also the changes of the environment, e.g. the presence of the object 900 in the vicinity of the first secondary electrode 401, can be detected. The second potential difference DV_2 can be arranged by connecting the first secondary electrode 401 to a second measurement potential V_{M2} , and conducting a second common potential V_{COM2} to the first primary electrode 301. The second potential difference DV_2 is, by definition (see above for DV_1), a difference ($V_{COM2} - V_{M2}$) between the potential of the first primary electrode 301 and the potential of the first secondary electrode 401. It is noted that the potential differences DV_1 and DV_2 are different from each other, even if their absolute values were the same. For example, both the common potentials V_{COM1} and V_{COM2} may be equal and both the measurement potentials V_{M1} and V_{M2} may also be equal. Moreover, both the common potentials V_{COM1} and V_{COM2} may be equal to a ground potential GND.

[0104] In this way, the secondary capacitance is indicative of at least one of: (i) a presence of an object in a vicinity of the first secondary electrode 401 and (ii) a property of material, the property affecting a dielectric constant of the material, such as humidity, heart rate, or sweating. Concerning the presence of an object 900, because the object 900 has a different dielectric constant than air, and may also be at least loosely connected to a ground potential, the object 900 affects the secondary capacitance C_{401} as measured by the first secondary electrode. In a similar manner, if the location of the object 900 is constant, humidifying the object 900 affects its dielectric constant, and thus also the secondary capacitance C_{401} . Moreover, if the object 900 is e.g. a hand or a finger of a person, the person's heart beat affects the dielectric constant, since a heartbeat temporarily increases the amount of blood in the finger/hand. Sweating also affects dielectric constant of the body part that sweats.

[0105] The first secondary electrode 401 can be used for the second functionality as described above even if only one wire W_{112} extends from the first secondary electrode 401, as shown in FIG. 1a. However, an electrode can be used in such a way, even if two wires W_{112} , W_{212} , extend from the first secondary electrode 401, as shown in FIG. 2e. This also shows that the electrode of FIG. 2e can be used for three different functionalities: measuring the primary capacitance C_{301} (indicative of compression), heating (by feeding current through the electrode), and measuring the secondary capacitance C_{401} (indicative of e.g. presence in a vicinity).

[0106] In an embodiment, the measuring of the secondary capacitance C_{401} does not comprise feeding electric current to the first secondary electrode 401 and at the same time receiving electric current from the first secondary electrode 401. This is contrast to enabling e.g. a heating functionality of first secondary electrode 401 of FIG. 2e, which involves feeding electric current to the first secondary electrode 401 and at the same time receiving electric current from the first secondary electrode 401 using the wires W_{112} , W_{212} .

[0107] Preferably, the arranging the second potential difference DV_2 comprises connecting to the first primary electrode 301 to a second common potential V_{COM2} that equals the first common potential V_{COM1} , to which during the first period of time the first secondary electrode 401 is connected. This has the effect that the common potential V_{COM2} , V_{COM1} is not, during the second period of time t_2 , shielding the first secondary electrode from the effects of the object 900. Moreover, during the first period of time t_1 , the common potential V_{COM2} , V_{COM1} shields the compressible layer 200 and thus also shields the effect of an object 900 on the primary capacitance C_{301} .

[0108] More preferably, the potential differences DV_2 and DV_1 have opposite signs. This is beneficial, since then the control unit 510 needs not generate both positive and negative voltages. In fact, only one operating voltage (different from the ground potential) suffices. When the potential differences DV_2 and DV_1 have opposite signs, their product is negative. Thus, in a preferable embodiment, a product $DV_1 \cdot DV_2$ of the first potential difference DV_1 and the second potential difference DV_2 is less than zero.

[0109] The second common potential V_{COM2} may e.g. be a ground potential GND. This has the effect that since the object 900 is typically at least loosely connected to a virtual electric ground G_p , the effect of the object 900 on the secondary capacitance C_{401} can more easily be sensed.

However, the potential of the virtual electric ground G_V need not be the same as the ground potential GND of the device 110.

[0110] In a corresponding manner, in an embodiment, the control unit 510 is configured to, during the second period of time t_2 , enable the second functionality by [i] arranging a second potential difference DV_2 between the first primary electrode 301 and the first secondary electrode 401, the second potential difference DV_2 being different from the first potential difference DV_1 and different from zero and [ii] measuring a secondary capacitance of the first secondary electrode 401. Preferably, the control unit 510 is configured to, during the second period of time t_2 , connect to the first primary electrode 301 to a second common potential V_{COM2} that equals the first common potential V_{COM1} .

[0111] More preferably, the control unit 510 is configured to, during the second period of time t_2 , connect to the first primary electrode 301 to a second common potential V_{COM2} that equals the first common potential V_{COM1} ; and connect the first secondary electrode 401 to a second measurement potential V_{M2} , such that a product $DV_1 \bullet DV_2$ of the first potential difference DV_1 and the second potential difference DV_2 is less than zero.

[0112] In an embodiment, during the second period of time t_2 , the device 110 is configured not to feed electric current to the first secondary electrode 401 and at the same time receive electric current from the first secondary electrode 401. In an embodiment, only one wire extends from the first secondary electrode 401.

[0113] This measurement principle has further benefits particularly when [a] the primary electrode layer 300 comprises at least a second primary electrode 302 and/or [b] the secondary electrode layer 400 comprises at least a second secondary electrode 402.

[0114] Concerning the option [a]: when the primary electrode layer 300 comprises at least a second primary electrode 302 (see FIGS. 1b, 1c, and 1d), in an embodiment, when the second functionality is enabled (during the second period of time t_2), the method comprises connecting the second primary electrode 302 to the same potential as the first primary electrode 301. In this way, the second primary electrode 302 and the first primary electrode 301 form a common electrode, relative to which the capacitance C_{401} is measured. Thus, the embodiment comprises measuring the secondary capacitance C_{401} of the first secondary electrode 401 relative to both the second primary electrode 302 and the first primary electrode 301 simultaneously. This has the effect that the secondary capacitance C_{401} can be measured more reliably, because the electrode relative to which it is measured (i.e. the combination of electrodes 301 and 302) is effectively larger than in the case these electrodes were not connected (in which case size of the reference electrode would be that of one of the electrodes 301 and 302).

[0115] Correspondingly, in an embodiment of the arrangement 100, the primary electrode layer 300 of the device 110 comprises at least a second primary electrode 302; and the control unit 510 is configured to, during the second period of time t_2 , connect the second primary electrode 302 to the same potential as the first primary electrode 301.

[0116] Concerning the option [b]: when the secondary electrode layer 400 comprises at least a second secondary electrode 402 (see FIGS. 1c, 1d, and 8), an embodiment comprises, measuring the capacitance C_{301} of the first primary electrode 301 (e.g. during the first period of time t_1)

such that while measuring the capacitance C_{301} , the second secondary electrode 402 is connected to the same potential as the first secondary electrode 401. This is depicted in FIG. 8. Thus, the second secondary electrode 402 and the first secondary electrode 401 in combination form a shield for shielding an electric field in compressible layer 200 from disturbances. Thus, the first and second secondary electrodes (401, 402) form a shield for the measuring of the capacitance C_{301} of the first primary electrode 301. For example, when measuring the compression caused by the object 900, the electrode 402 shields the electric field of the compressible layer 200 from the disturbance of the other object 910 (see FIG. 8). It is noted that this effect is beneficial also in the case the second functionality is a result of a transducer. Naturally the device 110 may comprise two electrodes 401, 402 in the secondary electrode layer 400, and further a component 490 (e.g. a transducer) arranged in or on the secondary electrode layer 400.

[0117] Thus, an embodiment of the method comprises, during the first period of time t_1 , connecting the second secondary electrode 402 to the first common potential V_{COM1} . Thus the capacitance C_{301} of the first primary electrode 301 is measured relative also to the second secondary electrode 402.

[0118] Correspondingly, in an embodiment of the arrangement 100, the secondary electrode layer 400 of the device 110 comprises at least a second secondary electrode 402; and the control unit 510 is configured to, during the first period of time t_1 , connect the second secondary electrode 402 to the same potential as the first secondary electrode 401.

[0119] Referring to FIGS. 7a to 7c, when the primary electrode layer 300 of the device 110 comprises several electrodes, several different capacitances can be measured, each capacitance corresponding to a compression of the compressible layer 200 at a location of that electrode. For example in FIG. 7a, a capacitance C_{301} of the first primary electrode 301 is indicative of compression of the compressible layer 200 in a vicinity of the first primary electrode 301, i.e. compression by the object 900. Moreover, a capacitance C_{302} of the second primary electrode 302 is indicative of compression of the compressible layer 200 in a vicinity of the second primary electrode 302. As shown in FIGS. 7a and 7b, the object 900 does not affect compression of the layer 200 in the vicinity of the electrode 302. Thus, to improve spatial accuracy of the device 110, the primary electrode layer 300 may be provided with more electrodes.

[0120] In an embodiment, the primary electrode layer 300 comprises at least a second primary electrode 302. Referring to FIG. 7b, the embodiment of the method comprises measuring the capacitance C_{302} of the second primary electrode 302. This capacitance C_{302} may be measured during the first period of time t_1 or during a third period of time. Preferably, while measuring the capacitance C_{302} of the second primary electrode 302 (i.e. during the first or third period of time), the second functionality is disabled. In the alternative, the effect of the second functionality can be taken into account in the measurement of the capacitance C_{302} in a similar manner as it can be taken into account in the measurement of the capacitance C_{301} mutatis mutandis.

[0121] For measuring the capacitance C_{302} , the method comprises connecting the first secondary electrode 401 and/or the second secondary electrode 402 to a potential, such as the first common potential V_{COM1} (as in FIG. 7b),

and conducting a potential $V_{M1,2}$ that is different from the potential of the first secondary electrode **401** and/or the potential of the second secondary electrode **402**, respectively, to the second primary electrode **302**. The potential $V_{M1,2}$ that is conducted to the second primary electrode **302** may, but need not, equal the first measurement potential V_{M1} (see FIGS. **7a** and **7b**). Referring to FIG. **7c**, during the second period of time **t2**, the second functionality can be enabled as discussed above.

[0122] Naturally, both the primary electrode layer **300** and the secondary electrode layer **400** may comprise multiple electrodes as detailed in FIG. **1c**. Thus, in an embodiment, the primary electrode layer **300** of the device **110** comprises the first primary electrode **301** and at least a second primary electrode **302**; and the secondary electrode layer **400** of the device **110** comprises the first secondary electrode **401** and at least a second secondary electrode **402**. These electrodes may be used as detailed above. For example, during the first period of time **t1**, the second functionality may be disabled, all the electrodes (**401**, **402**) of the secondary electrode layer **400** may be connected to a same potential, and capacitances C_{301} , C_{302} of each electrode (**301**, **302**) of the primary electrode layer **300** may be measured subsequently. However, the electrodes **401**, **402** of the secondary electrode layer need not be connected to the same potential. Preferably, the capacitance C_{302} of the second primary electrode **302** is measured relative to at least the second secondary electrode **402**. Thus, an embodiment comprises

[0123] connecting the second secondary electrode **402** to a potential, such as the first common potential V_{COM1} ,

[0124] conducting a potential that is different from the potential of the second secondary electrode **402** to the second primary electrode **302**, and

[0125] measuring a capacitance C_{302} of the second primary electrode **302** relative to the second secondary electrode **402** for determining another compression of the layer **200** of compressible material.

[0126] Correspondingly, in an embodiment, the control unit **510** of the arrangement **100** is configured to connect the second secondary electrode **402** to a potential, such as the first common potential V_{COM1} ; conduct a potential that is different from the potential of the second secondary electrode **402** to the second primary electrode **302**; and measure a capacitance C_{302} of the second primary electrode **302** relative to (at least) the second secondary electrode **402** for determining another compression of the layer **200** of compressible material.

[0127] Preferably, at the time of measuring a capacitance C_{302} of the second primary electrode **302** relative to the second secondary electrode **402**, the second functionality is disabled. However, as detailed above, this need not be the case, because the effect of the second functionality can be taken into account as detailed above. The capacitance C_{302} of the second primary electrode **302** relative to the second secondary electrode **402** may be measured during the first period of time **t1**, or at the third period of time, or at a fourth period of time.

[0128] During the second period of time **t2**, the second functionality is enabled, and all the electrodes (**301**, **302**) may be connected to a same potential. Preferably they are connected to a same potential when the second functionality is sensitive to variation in electric fields, e.g. when the second functionality comprises measuring a capacitance C_{401} , C_{402} of an electrode of the secondary electrode layer.

However, the electrodes **301**, **302** of the primary electrode layer **300** need not be connected to a same potential, because the second functionality may be insensitive to electric fields. The electrodes **401**, **402** of the secondary electrode layer **400** and/or the component **490** of the secondary electrode layer **400** may be operated as detailed above.

[0129] As indicated above, an embodiment of a method for operating the layered device **110** comprises during a second period of time **t2** [A] detecting that the second functionality is enabled, [B] enabling the second functionality of the device **110** by arranging a second potential difference DV_2 between the first primary electrode **301** and the first secondary electrode **401**, the second potential difference DV_2 being different from the first potential difference DV_1 and different from zero, [C] activating a component **490** of the layered device **110**, and/or [D] arranging an object to a galvanic contact with the first secondary electrode **401**.

[0130] Concerning the option [B], the second functionality need not be capacitive sensing of presence of an object in the vicinity of the first secondary electrode **401**. In an embodiment, the second functionality comprises another functionality than detecting presence of an object in the vicinity of the first secondary electrode **401**. In an embodiment, the second functionality comprises another functionality than detecting presence of an object in the vicinity of any electrode of the secondary electrode layer **400**. In addition or alternatively, in an embodiment, the second functionality comprises another functionality than measuring a quantity that is measurable from the same signal from which the capacitance C_{301} between the first primary electrode **301** and the first secondary electrode **401** is measured to determine the compression of the compressible layer **200**. In an embodiment, the second functionality comprises another functionality than measuring a quantity that is measurable from such a signal that is used to measure a capacitance between an electrode of the first primary electrode layer **300** and an electrode of the secondary electrode layer **400** for detecting the compression of the compressible layer **200**.

[0131] In an embodiment, between the primary electrode layer **300** and the secondary electrode layer **400** only dielectric material has been arranged. Thus, preferably, the compressible layer **200** does not comprise an electrically at least semiconducting layer arranged in between dielectric layers. This has the effect that the structure remains simple. In particular, in an embodiment, the device **110** does not comprise electrically at least semiconducting material between the first primary electrode **301** and the first secondary electrode **401**. Herein an electrically at least semiconducting material has a resistivity less than 10 m at a temperature of 23°C .

[0132] However, referring to FIG. **1d**, an accuracy of the measured degree of compression can be improved by arranging the first primary electrode **301** (or the whole primary electrode layer **300**) in between the compressible layer **200** and a secondary compressible layer **610**; and arranging at least a part of the secondary compressible layer **610** between the first primary electrode **301** (or the whole primary electrode layer) and an electrical conductor **710**. What has been said about the properties of the compressible layer **200** applies to the secondary compressible layer **610**. Thus, the first primary electrode **301** forms two capacitances: a first capacitance relative to at least the first secondary electrode **401** and a second capacitance relative to

the electrical conductor 710. These may be measured separately following the principles detailed above. In the alternative, the first primary electrode 301 forms the capacitance C_{301} relative to a combination of the electrical conductor 710 and at least the first secondary electrode 401. The scale of the compression sensor can also be engineered for different purposes e.g. by using different material in the secondary compressible layer 610 than in the compressible layer 200.

[0133] It has been found that when the second functionality involves a measurement of the capacitance of an electrode (401, 402) of the secondary electrode layer, an electrically conducting or semiconducting layer in between the object 900 and the electrode 401, 402 would reduce sensitivity of the capacitive measurement. Moreover, naturally, if the second functionality involves emitting light, an opaque layer should not be applied on the light emitting component.

[0134] For these reasons, in an embodiment, the secondary electrode layer 400 is not arranged between the compressible layer 200 and such a layer that is electrically at least semiconducting and opaque. In an embodiment, the first secondary electrode 401 is not arranged between the compressible layer 200 and such a layer that is electrically conducting. In an embodiment, a component 490 is not arranged between the compressible layer 200 and such a layer that is opaque. As an example, FIG. 2b shows the second covering 820, and the secondary electrode layer 400 is arranged between the second covering 820 and the compressible layer 200. If the second covering 820 was opaque or electrically at least semiconducting, the second covering could effectively make the second functionality unusable.

[0135] However, referring to FIG. 1e, the primary electrode layer 300 may be covered with a first covering 810. Moreover, the secondary electrode layer 400 may be covered with a second covering 820. As detailed above, preferably, if a second covering 820 is applied, the second covering 820 is preferably transparent or electrically insulating. For definition of electrically insulating material, reference is made to what has been said above. As for transparency, solely for purposes of clarity, in this context a transmittivity of transparent material in the visual spectrum of radiation (from 400 nm to 750 nm) is at least 25% at a temperature of 23° Celsius.

[0136] The device 110 measures the capacitance well at least when the electrode (e.g. 301 or 302) that is used for measuring the capacitance at least partially overlaps an electrode (e.g. 401 or 402, respectively) relative to which the capacitance is measured. Herein the meaning of overlapping is being on top of each other in a direction that is parallel to a direction Sz of thickness of the device at the location in which the occurrence of overlapping or non-overlapping is determined. It is noted that since the device 110 may be conformable, and may be made to conform a non-planar object, a direction of thickness of the device depends on location. In any case, the direction of thickness is unidirectional with a norm of the primary electrode layer 300 (see Figs.)

[0137] Thus, in a preferable embodiment, at least a part of the first primary electrode 301 overlaps with at least a part of the first secondary electrode 401. More specifically, in a preferable embodiment, at least a part of the first primary electrode 301 overlaps, in a direction that is parallel to a direction of a thickness of the device 110, with at least a part

of the first secondary electrode 401; wherein a direction of a thickness of the device 110 is parallel to a norm of the primary electrode layer 300. When the device 110 comprises the electrodes 301, 302, 401, and 402 as discussed above, preferably, at least a part of the second primary electrode 302 overlaps, in a direction that is parallel to a direction of a thickness of the device 110, with at least a part of the second secondary electrode 402.

[0138] More preferably, the part of the first primary electrode 301 that overlaps with the at least a part of the first secondary electrode 401 has an overlapping area such that the overlapping area is at least 50% of an area of the smaller of the first primary electrode 301 and the first secondary electrode 401. When the device 110 comprises the electrodes 301, 302, 401, and 402 as discussed above, preferably also the part of the second primary electrode 302 that overlaps with the at least a part of the second secondary electrode 402 has an overlapping area such that the overlapping area is at least 50% of an area of the smaller of the second primary electrode 302 and the second secondary electrode 402.

[0139] When measuring the capacitance C_{301} of the first primary electrode 301, preferably, a capacitance of the whole first primary electrode 301 is measured by one measurement. Therefore, preferably, the first primary electrode 301 does not overlap at least two secondary electrodes (e.g. 401, 402) of the secondary electrode layer 400 such that, when measuring the capacitance C_{301} of the first primary electrode 301, the electrodes of the secondary electrode layer that the first primary electrode 301 overlaps are arranged at different potentials. Preferably, the first primary electrode 301 does not overlap at least two secondary electrodes (e.g. 401, 402) of the secondary electrode layer 400. However, if the first primary electrode 301 overlaps at least two secondary electrodes (e.g. 401, 402) of the secondary electrode layer 400, if these secondary electrodes are arranged at the same potential, they form a common reference electrode, as discussed above, and the capacitance of the whole first primary electrode can be measured by one measurement.

[0140] Correspondingly, in an embodiment, the control unit 510 is configured to measure a capacitance from the whole area of the first primary electrode 301. In an embodiment, the control unit 510 is configured to measure a capacitance from the whole area of the first primary electrode 301 at one instance of time. In an embodiment, the control unit 510 is configured to measure a capacitance from the whole area of each one of the second primary electrode 302, too. By measuring the capacitance of a whole area of the electrode(s), there is no need, e.g. for measuring different parts of the electrodes in sequence, which improves the temporal accuracy of the measurements. More specifically, in an embodiment, the controller 500 is configured to measure: the capacitance from the whole area of the first primary electrode 301 at one instance of time and the capacitance from the whole area of the second primary electrode 302 at one instance of time.

[0141] In the above, the method, device and arrangement have been discussed mainly in connection with embodiments, wherein compressible and electrically insulating material 200m (see FIG. 9) has been arranged, as a compressible layer 200, in between the primary electrode layer 300 and the secondary electrode layer 400.

[0142] However, referring to FIG. 9, the first primary electrode 301 need not be comprised by a primary electrode

layer 300. Instead, the first primary electrode 301 may be e.g. an electrode formed by sewing with a conductive yarn directly on the compressible and electrically insulating material 200m. In a similar manner the first secondary electrode 401 need not be comprised by a secondary electrode layer 400. The first secondary electrode 401 may be e.g. an electrode formed by sewing with a conductive yarn directly on the compressible and electrically insulating material 200m. As for measuring the capacitance, it suffices that compressible and electrically insulating material 200m is arranged between the first primary electrode 301 and the first secondary electrode 401 irrespective of any of: [a] that the compressible and electrically insulating material 200m is comprised by a compressible layer, [b] that the first primary electrode 301 is comprised by a primary electrode layer 300, and/or [c] that the first secondary electrode 401 is comprised by a secondary electrode layer 400.

[0143] However, when the device 110 comprises the second primary electrode 302, the second primary electrode 302 and the first primary electrode 301 may be arranged relative to each other on a same plane (e.g. a first plane, see e.g. FIG. 7c). In a similar manner, when the device 110 comprises the second secondary electrode 302, the second secondary electrode 402 and the first secondary electrode 401 may be arranged relative to each other on a same plane (e.g. a second plane, see e.g. FIG. 8). The compressible and electrically insulating material 200m may be arranged between these planes (i.e. the first plane and the second plane).

[0144] Concerning the placement of the component 490, it suffices that the device 110 comprises the component 490. However, preferably, the component is arranged in one of the following locations:

- [0145] in or on the primary electrode layer 300 (if present in the device 110),
- [0146] in or on the secondary electrode layer 400 (if present in the device 110),
- [0147] in or on the compressible and electrically insulating material 200m, or
- [0148] in or on the compressible layer 200 comprising the compressible and electrically insulating material 200m.

EXAMPLES SUMMARIZING SOME EMBODIMENTS

- [0149] 1. A method for operating a device (110) having at least a first functionality and a second functionality,
- [0150] the first functionality being capacitive sensing of compression and
- [0151] the second functionality being another, different, functionality; and for enabling the first functionality the device (110) comprises
- [0152] a first primary electrode (301),
- [0153] a first secondary electrode (401), and
- [0154] compressible and electrically insulating material (200m) between the first primary electrode (301) and the first secondary electrode (401);
- [0155] the method comprising during a first period of time
- [0156] arranging a first potential difference (DV₁) between the first primary electrode (301) and the first secondary electrode (401) by
- [0157] connecting the first secondary electrode (401) to a first common potential (V_{COM1}), and

[0158] conducting a first measurement potential (V_{M1}) to the first primary electrode (301), the first measurement potential (V_{M1}) being different from the first common potential (V_{COM1}), whereby

[0159] the first potential difference (DV₁) is a difference (V_{M1}-V_{COM1}) between the first measurement potential (V_{M1}) and the first common potential (V_{COM1}) and

[0160] measuring a capacitance (C₃₀₁) of the first primary electrode (301) relative to the first secondary electrode (401) for determining a compression of the compressible and electrically insulating material (200m),

[0161] the method comprising

[0162] detecting that the second functionality is enabled during a second period of time or

[0163] during a second period of time, enabling the second functionality, wherein

[0164] the second functionality is enabled by using the first primary electrode (301), the first secondary electrode (401), or a component (490) of the device (110), and

[0165] the second functionality comprises another functionality than capacitive measurement of a quantity such that the first secondary electrode (401) is connected to the first common potential (V_{COM1}), and the first measurement potential (V_{M1}) is conducted to the first primary electrode (301).

[0166] 2. The method of Example 1, wherein the device (100) comprises

[0167] the component (490), the component (490) optionally being the first primary or secondary electrode (301, 401) and

[0168] a primary wire electrically connected to the component (490);

[0169] and in the method

[0170] the second functionality comprises transducing, by using the component (490), (i) an electric current or voltage to a non-electric signal or (ii) a non-electric signal to an electric current or voltage;

[0171] preferably,

[0172] the second functionality involves feeding an electric current to the component (490) or receiving an electric current from the component (490);

[0173] more preferably,

[0174] the component (490) is a heater, an antenna, a light emitter, a display, or a sensor, such as a temperature sensor, a vital signs sensor, an inductive sensor, a piezoresistive sensor, or a strain gauge.

[0175] 3. The method of Example 2, wherein the device (100) comprises

[0176] a secondary wire electrically connected to the component (490);

[0177] and in the method

[0178] the second functionality involves, e.g. the second functionality is enabled by, feeding an electric current to the component (490) and receiving an electric current from the component (490);

[0179] more preferably,

[0180] the component (490) is a heater, a light emitter, a display, or a sensor, such as a vital signs sensor, an inductive sensor, a piezoresistive sensor, or a strain gauge.

- [0181] 4. The method of any of the Examples 1 to 3, wherein the method comprising, during the second period of time, enabling the second functionality, wherein
- [0182] enabling the second functionality comprises
- [0183] arranging a second potential difference (DV_2) between the first primary electrode (301) and the first secondary electrode (401), the second potential difference (DV_2) being different from the first potential difference (DV_1) and different from zero and
- [0184] measuring a secondary capacitance of the first secondary electrode (401);
- [0185] preferably,
- [0186] the secondary capacitance is indicative of at least one of: (i) a presence of an object in a vicinity of the first secondary electrode (401), and (ii) a property of material, the property affecting a dielectric constant of the material, such as humidity;
- [0187] more preferably,
- [0188] the arranging of the second potential difference (DV_2) comprises connecting to the first primary electrode (301) to a second common potential (V_{COM2}) that equals the first common potential (V_{COM1}), preferably such that a product ($DV_1 \bullet DV_2$) of the first potential difference (DV_1) and the second potential difference (DV_2) is less than zero.
- [0189] 5. The method of any of the Examples 1 to 4, wherein
- [0190] the device (110) comprises at least a second primary electrode (302), and
- [0191] enabling the second functionality comprises connecting the second primary electrode (302) to the same potential as the first primary electrode (301); preferably,
- [0192] the second primary electrode (302) and the first primary electrode (301) are arranged on a first plane;
- [0193] more preferably,
- [0194] in the embodiment of Example 4, the secondary capacitance of the first secondary electrode (401) is measured relative to both the second primary electrode (302) and the first primary electrode (301) because of the connecting the second primary electrode (302) to the same potential as the first primary electrode (301).
- [0195] 6. The method of any of the Examples 1 to 5, wherein
- [0196] [A]
- [0197] during the first period of time (t_1), the device (110) is arranged between a support (990) and an object (900) causing the compression of the compressible material (200m) such that the first primary electrode (301) is closer to the support (990) than the first secondary electrode (401), and
- [0198] during the second period of time (t_2), the device (110) is arranged such that the first primary electrode (301) is closer to the support (990) than the first secondary electrode (401) OR
- [0199] [B]
- [0200] during the first period of time (t_1), the device (110) is arranged between a support (990) and an object (900) causing the compression of the compressible material (200m) such that the first secondary electrode (401) is closer to the support (990) than the first primary electrode (301), and
- [0201] during the second period of time (t_2), the device (110) is arranged such that the first secondary electrode (401) is closer to the support (990) than the first primary electrode (301).
- [0202] 7. The method of any of the Examples 1 to 6, wherein the method comprises, during the second period of time, enabling the second functionality, wherein
- [0203] enabling the second functionality comprises controlling the second functionality such that during at least a part of the first period of time the second functionality is deactivated and during the second period of time the second functionality is activated.
- [0204] 8. The method of any of the Examples 1 to 7, wherein the method comprises, detecting that the second functionality is enabled during the second period of time, wherein detecting that the second functionality is enabled comprises
- [0205] measuring a signal from a primary electrode layer or from a secondary electrode layer during the second period of time,
- [0206] determining, from the signal, that the second functionality has been enabled, and
- [0207] compensating for the effect of the second functionality on the capacitance of the first primary electrode or disregarding the capacitance measured during the second period of time;
- [0208] preferably, the method further comprises
- [0209] determining, from the capacitance measured during the first period of time, that the second functionality has been disabled during at least a part of the first period of time and/or
- [0210] measuring a signal from a secondary electrode layer during the first period of time and determining, from the signal, that the second functionality has been disabled.
- [0211] 9. The method of any of the Examples 1 to 8, wherein
- [0212] the device (110) comprises at least a second secondary electrode (402); and the method comprises during the first period of time
- [0213] connecting the second secondary electrode (402) to the first common potential (V_{COM1}), whereby
- [0214] the capacitance (C_{301}) of the first primary electrode (301) is measured relative also to the second secondary electrode (402);
- [0215] preferably,
- [0216] the second secondary electrode (402) and the first secondary electrode (401) are arranged on a second plane;
- [0217] more preferably,
- [0218] the method comprises enabling the second functionality and the enabling the second functionality comprises
- [0219] arranging a second potential difference (DV_2) between the first primary electrode (301) and the first secondary electrode (401), the second potential difference (DV_2) being different from the first potential difference (DV_1) and different from zero and
- [0220] measuring a secondary capacitance of the first secondary electrode (401);

- [0221] more preferably,
- [0222] the arranging a second potential difference (DV_2) comprises connecting to the first primary electrode (301) to a second common potential (V_{COM2}) that equals the first common potential (V_{COM1}), preferably such that a product ($DV_1 \bullet DV_2$) of the first potential difference (DV_1) and the second potential difference (DV_2) is less than zero.
- [0223] 10. The method of any of the Examples 1 to 9, wherein
- [0224] the device (110) comprises at least a second primary electrode (302), and the method comprises
- [0225] connecting the first secondary electrode (401) to a potential, such as the first common potential (V_{COM1}),
- [0226] conducting a potential that is different from the potential of the first secondary electrode (401) to the second primary electrode (302), and
- [0227] measuring a capacitance (C_{302}) of the second primary electrode (302) relative to the first secondary electrode (401) or a second secondary electrode (402) for determining another compression of a layer comprising the compressible material (200m);
- [0228] preferably,
- [0229] the second primary electrode (302) and the first primary electrode (301) are arranged on a first plane.
- [0230] 11. An arrangement (100) for operating a device (110) having at least a first functionality and a second functionality,
- [0231] the first functionality being capacitive sensing of compression and
- [0232] the second functionality being another, different, functionality; the device (110) comprising
- [0233] a first primary electrode (301),
- [0234] a first secondary electrode (401), and
- [0235] compressible and electrically insulating material (200m) between the primary electrode (301) and the secondary electrode (401); the arrangement (100) comprising a control unit (510) configured to
- [0236] during a first period of time,
- [0237] arrange a first potential difference (DV_1) between the first primary electrode (301) and the first secondary electrode (401) by
- [0238] connecting the first secondary electrode (401) to a first common potential (V_{COM1}), and
- [0239] conducting a first measurement potential (V_{M1}) to the first primary electrode (301), the first measurement potential (V_{M1}) being different from the first common potential (V_{COM1}), whereby
- [0240] the first potential difference (DV_1) is a difference ($V_{M1} - V_{COM1}$) between the first measurement potential (V_{M1}) and the first common potential (V_{COM1}) and
- [0241] measure a capacitance (C_{301}) of the first primary electrode (301) relative to the first secondary electrode (401) for determining a compression of the compressible and electrically insulating material (200m), and
- [0242] control unit (510) is configured to
- [0243] detect that the second functionality of the device (110) is enabled during a second period of time or
- [0244] enable the second functionality of the device (110) at a second period of time, wherein
- [0245] the second functionality is configured to be enabled by using the first primary electrode (301), the first secondary electrode (401), or a component (490) of the device (110), and
- [0246] the second functionality comprises another functionality than capacitive measurement of a quantity such that the first secondary electrode (401) is connected to the first common potential (V_{COM1}), and the first measurement potential (V_{M1}) is conducted to the first primary electrode (301).
- [0247] 12. The arrangement (100) of Example 11, wherein the device (110) comprises
- [0248] a component (490) configured to enable the second functionality, the component (490) optionally being the first primary or secondary electrode (301, 401),
- [0249] a primary wire electrically connecting the component (490) to the control unit (510), and
- [0250] a secondary wire electrically connecting the component (490) to the control unit (510);
- [0251] preferably,
- [0252] the component (490) is a transducer configured to transduce (i) an electric current or voltage to a non-electric signal or (ii) a non-electric signal to an electric current or voltage;
- [0253] more preferably,
- [0254] the component (490) is a heater, a light emitter, a display, or a sensor, such as a vital signs sensor, an inductive sensor, a piezoresistive sensor, or a strain gauge.
- [0255] 13. The arrangement (100) of Example 11 or 12, wherein the control unit (510) is configured to, during the second period of time (t_2), enable the second functionality by
- [0256] arranging a second potential difference (DV_2) between the first primary electrode (301) and the first secondary electrode (401), the second potential difference (DV_2) being different from the first potential difference (DV_1) and different from zero and
- [0257] measuring a secondary capacitance of the first secondary electrode (401);
- [0258] preferably,
- [0259] the control unit (510) is configured to, during the second period of time (t_2), connect to the first primary electrode (301) to a second common potential (V_{COM2}) that equals the first common potential (V_{COM1}),
- [0260] more preferably,
- [0261] a product ($DV_1 \bullet DV_2$) of the first potential difference (DV_1) and the second potential difference (DV_2) is less than zero.
- [0262] 14. The arrangement of any of the Examples 11 to 13, wherein
- [0263] the control unit (510) is configured to control the second functionality such that during at least a part of the first period of time the second functionality is deactivated and during the second period of time the second functionality is activated.
- [0264] 15. The arrangement of any of the Examples 11 to 14, wherein the control unit (510) is configured to
- [0265] measure a signal indicative of the second functionality being enabled,

- [0266] determine, from the signal, that the second functionality has been enabled, and
- [0267] compensate for the effect of the second functionality on the capacitance of the first primary electrode or disregard the capacitance measured during the second period of time;
- [0268] preferably, the control unit (510) is configured to
- [0269] determine, from the capacitance measured during the first period of time, that the second functionality has been disabled during the at least a part of the first period of time and/or
- [0270] measure a signal from a secondary electrode layer during the first period of time and determining, from the signal, that the second functionality has been disabled.
- [0271] 16. The method of any of the Examples 1 to 10 or the arrangement of any of the Examples 11 to 15, wherein
- [0272] the first secondary electrode (401) is not arranged between the compressible and electrically insulating material and such a layer that is electrically at least semiconducting and opaque;
- [0273] preferably,
- [0274] the first secondary electrode (401) is not arranged between the compressible and electrically insulating material and such a layer that is electrically conducting.
- [0275] 17. The method of any of the Example 1 to 10 or 16 or the arrangement of any of the Example 11 to 16, wherein
- [0276] at least a part of the first primary electrode (301) overlaps with at least a part of the first secondary electrode (401);
- [0277] preferably,
- [0278] the part of the first primary electrode (301) the overlaps with the at least a part of the first secondary electrode (401) has an overlapping area such that
- [0279] the overlapping area is at least 50% of an area of the smaller of the first primary electrode (301) and the first secondary electrode (401).
- [0280] 18. The method of any of the Examples 1 to 10 or 16 or 17 or the arrangement of any of the Examples 11 to 17, wherein
- [0281] at least a part of at least one of the component (490), the first primary electrode (301), and the first secondary electrode (401) is exposed so that
- [0282] an object can be arranged in galvanic contact with the component (490), the first primary electrode (301), and/or the first secondary electrode (401) without removing solid material from the device (110) or penetrating through solid material of the device (110).
- [0283] 19. The method of any of the Examples 1 to 10 or 17 or 18 or the arrangement of any of the Examples 11 to 18, wherein
- [0284] the device (110) is reversibly stretchable by at least 10% at a temperature 20 C;
- [0285] preferably,
- [0286] the device (110) is conformable so that at least a part thereof is arranged to conform a surface of a hemisphere having a radius of 10 cm at a temperature of 20° C. without introducing plastic deformations to the materials of the device (110).
1. A method for operating a device having at least a first functionality and a second functionality,
- the first functionality being capacitive sensing of compression; and
- the second functionality being another, different, functionality; and for enabling the first functionality the device comprises:
- a first primary electrode,
- a first secondary electrode, and
- compressible and electrically insulating material between the first primary electrode and the first secondary electrode;
- the method comprising during a first period of time:
- arranging a first potential difference between the first primary electrode and the first secondary electrode by: connecting the first secondary electrode to a first common potential, and
- conducting a first measurement potential to the first primary electrode, the first measurement potential being different from the first common potential, whereby
- the first potential difference is a difference between the first measurement potential and the first common potential, and
- measuring a capacitance of the first primary electrode relative to the first secondary electrode for determining a compression of the compressible and electrically insulating material,
- the method comprising:
- detecting that the second functionality is enabled during a second period of time, or
- during a second period of time, enabling the second functionality, wherein:
- the second functionality is enabled by using the first secondary electrode,
- the second functionality comprises another functionality than capacitive measurement of a quantity such that the first secondary electrode is connected to the first common potential and the first measurement potential is conducted to the first primary electrode,
- during the first period of time, the device is arranged between a support and an object causing compression of the compressible material such that the first primary electrode is closer to the support than the first secondary electrode, and
- during the second period of time, the device is arranged such that the first primary electrode is closer to the support than the first secondary electrode.
2. The method of claim 1, wherein the device comprises: a primary wire electrically connected to the first secondary electrode;
- and wherein:
- the second functionality comprises transducing, by using first secondary electrode, an electric current or voltage to a non-electric signal or a non-electric signal to an electric current or voltage;
- preferably,
- the second functionality involves feeding an electric current to the first secondary electrode or receiving an electric current from the first secondary electrode;
- more preferably,
- the first secondary electrode is a heater, an antenna, a temperature sensor, a vital signs sensor, an inductive sensor, a piezoresistive sensor, or a strain gauge.

3. The method of claim 2, wherein the device comprises: a secondary wire electrically connected to the first secondary electrode; and wherein:
the second functionality involves, e.g. the second functionality is enabled by, feeding an electric current to the first secondary electrode and receiving an electric current from the first secondary electrode;
more preferably,
the first secondary electrode is a heater, a vital signs sensor, an inductive sensor, a piezoresistive sensor, or a strain gauge.
4. The method of claim 1, wherein the method comprising, during the second period of time, enabling the second functionality, wherein enabling the second functionality comprises:
arranging a second potential difference between the first primary electrode and the first secondary electrode, the second potential difference being different from the first potential difference and
measuring a secondary capacitance of the first secondary electrode;
preferably,
the secondary capacitance is indicative of at least one of: a presence of an object in a vicinity of the first secondary electrode, and a property of material, the property affecting a dielectric constant of the material, such as humidity;
more preferably,
the arranging of the second potential difference comprises connecting to the first primary electrode to a second common potential that equals the first common potential, preferably such that a product of the first potential difference and the second potential difference is less than zero.
5. The method of claim 1, wherein
the device comprises at least a second primary electrode, and
enabling the second functionality comprises connecting the second primary electrode to a same potential as the first primary electrode; preferably,
the second primary electrode and the first primary electrode are arranged on a first plane.
6. The method of claim 1, wherein the method comprises, during the second period of time, enabling the second functionality, wherein
enabling the second functionality comprises controlling the second functionality such that during at least a part of the first period of time the second functionality is deactivated and during the second period of time the second functionality is activated.
7. The method of claim 1, wherein the method comprises, detecting that the second functionality is enabled during the second period of time, wherein detecting that the second functionality is enabled comprises:
measuring a signal from a primary electrode layer or from a secondary electrode layer during the second period of time,
determining, from the signal, that the second functionality has been enabled, and
compensating for an effect of the second functionality on the capacitance of the first primary electrode or disregarding the capacitance measured during the second period of time;
- preferably, the method further comprises:
determining, from the capacitance measured during the first period of time, that the second functionality has been disabled during at least a part of the first period of time and/or
measuring a signal from a secondary electrode layer during the first period of time and determining, from the signal, that the second functionality has been disabled.
8. The method of claim 1, wherein
the device comprises at least a second secondary electrode; and the method comprises during the first period of time,
connecting the second secondary electrode to the first common potential, whereby the capacitance of the first primary electrode is measured relative also to the second secondary electrode;
preferably,
the second secondary electrode and the first secondary electrode are arranged on a second plane;
more preferably,
the method comprises enabling the second functionality and the enabling the second functionality comprises:
arranging a second potential difference between the first primary electrode and the first secondary electrode, the second potential difference being different from the first potential difference and
measuring a secondary capacitance of the first secondary electrode;
more preferably,
the arranging a second potential difference comprises connecting to the first primary electrode to a second common potential that equals the first common potential, preferably such that a product of the first potential difference and the second potential difference is less than zero.
9. The method of claim 1, wherein
the device comprises at least a second primary electrode, and the method comprises:
connecting the first secondary electrode to a potential, such as the first common potential,
conducting a potential that is different from the potential of the first secondary electrode to the second primary electrode, and
measuring a capacitance of the second primary electrode relative to the first secondary electrode or a second secondary electrode for determining another compression of a layer comprising the compressible material;
preferably,
the second primary electrode and the first primary electrode are arranged on a first plane.
10. An arrangement for operating a device having at least a first functionality and a second functionality,
the first functionality being capacitive sensing of compression, and
the second functionality being another, different, functionality; the device comprising:
a first primary electrode,
a first secondary electrode, and
compressible and electrically insulating material between the primary electrode and the secondary electrode; the arrangement comprising a control unit configured to:

during a first period of time,
 arrange a first potential difference between the first primary electrode and the first secondary electrode by connecting the first secondary electrode to a first common potential, and
 conducting a first measurement potential to the first primary electrode, the first measurement potential being different from the first common potential, whereby
 the first potential difference is a difference between the first measurement potential and the first common potential and
 measure a capacitance of the first primary electrode relative to the first secondary electrode for determining a compression of the compressible and electrically insulating material, and
 control unit is configured to;
 detect that the second functionality of the device is enabled during a second period of time or
 enable the second functionality of the device at a second period of time, wherein:
 the second functionality is configured to be enabled by using the first secondary electrode,
 the second functionality comprises another functionality than capacitive measurement of a quantity such that the first secondary electrode is connected to the first common potential and the first measurement potential is conducted to the first primary electrode; the arrangement comprising:
 a support such that that the first primary electrode is closer to the support than the first secondary electrode.

11. The arrangement of claim 10, wherein the device comprises
 a primary wire electrically connecting the first secondary electrode to the control unit, and
 a secondary wire electrically connecting the first secondary electrode to the control unit;
 preferably,
 the first secondary electrode is configured to transduce an electric current or voltage to a non-electric signal or a non-electric signal to an electric current or voltage;
 more preferably,
 the first secondary electrode is a heater, a vital signs sensor, an inductive sensor, a piezoresistive sensor, or a strain gauge.

12. The arrangement of claim 10, wherein the control unit is configured to, during the second period of time, enable the second functionality by:
 arranging a second potential difference between the first primary electrode and the first secondary electrode, the second potential difference being different from the first potential difference, and
 measuring a secondary capacitance of the first secondary electrode;
 preferably,
 the control unit is configured to, during the second period of time, connect to the first primary electrode to a second common potential that equals the first common potential,

more preferably,
 a product of the first potential difference and the second potential difference is less than zero.

13. The arrangement of claim 10, wherein the control unit is configured to control the second functionality such that during at least a part of the first period of time the second functionality is deactivated, and during the second period of time the second functionality is activated.

14. The arrangement of claim 10, wherein the control unit is configured to;

measure a signal indicative of the second functionality being enabled,

determine, from the signal, that the second functionality has been enabled, and

compensate for the effect of the second functionality on the capacitance of the first primary electrode or disregard the capacitance measured during the second period of time;

preferably, the control unit is configured to:

determine, from the capacitance measured during the first period of time, that the second functionality has been disabled during at least a part of the first period of time, and/or

measure a signal from a secondary electrode layer during the first period of time and determining, from the signal, that the second functionality has been disabled.

15. The method of claim 1, wherein

the first secondary electrode is not arranged between the compressible and electrically insulating material and a layer that is electrically at least semiconducting and opaque;

preferably,

the first secondary electrode is not arranged between the compressible and electrically insulating material and a layer that is electrically conducting.

16. The arrangement of claim 10, wherein:

the first secondary electrode is not arranged between the compressible and electrically insulating material and a layer that is electrically at least semiconducting and opaque;

preferably,

the first secondary electrode is not arranged between the compressible and electrically insulating material and a layer that is electrically conducting.

17. The method of claim 4, wherein:

the device comprises at least a second primary electrode, enabling the second functionality comprises connecting the second primary electrode to a same potential as the first primary electrode, and

the secondary capacitance of the first secondary electrode is measured relative to both the second primary electrode and the first primary electrode because of the connecting the second primary electrode to the same potential as the first primary electrode.

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